

Data Structures and Algorithms I.

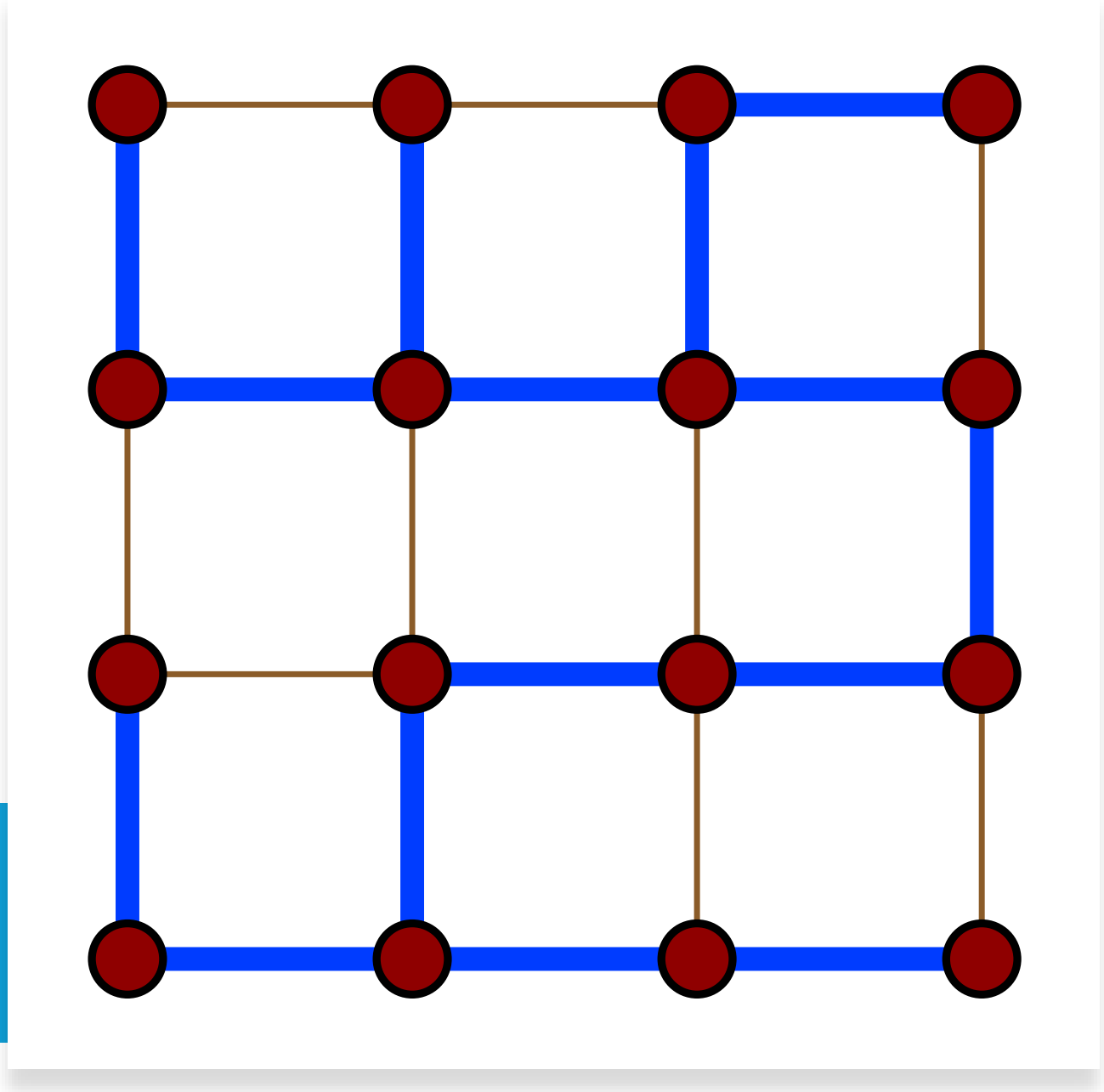
University of Szeged – Robert Nagy

2025/26. Spring Semester

Minimum spanning tree

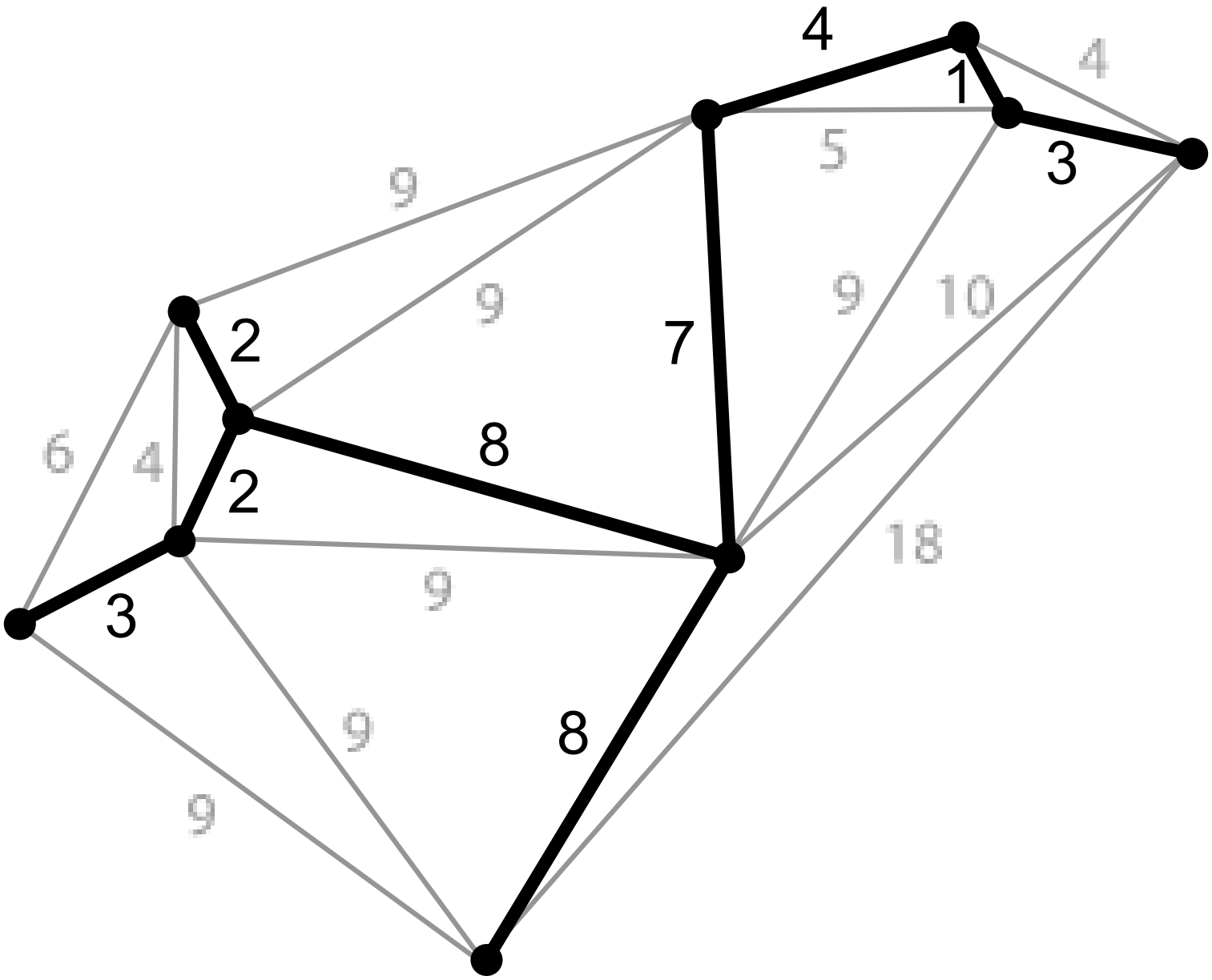
- Weighted, undirected graph

https://en.wikipedia.org/wiki/Spanning_tree#/media/File:4x4_grid_spanning_tree.svg



Spanning tree

https://en.wikipedia.org/wiki/Minimum_spanning_tree#/media/File:Minimum_spanning_tree.svg



Real world usage



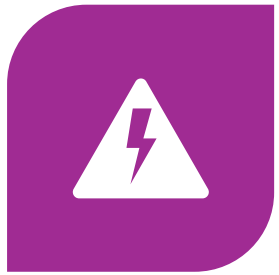
TELECOMMUNICATION NETWORKS



TRANSPORTATION NETWORKS



WATER SUPPLY NETWORKS



ELECTRICAL GRIDS

Kruskal’s algorithm

1. Sort the edges

Kruskal’s algorithm

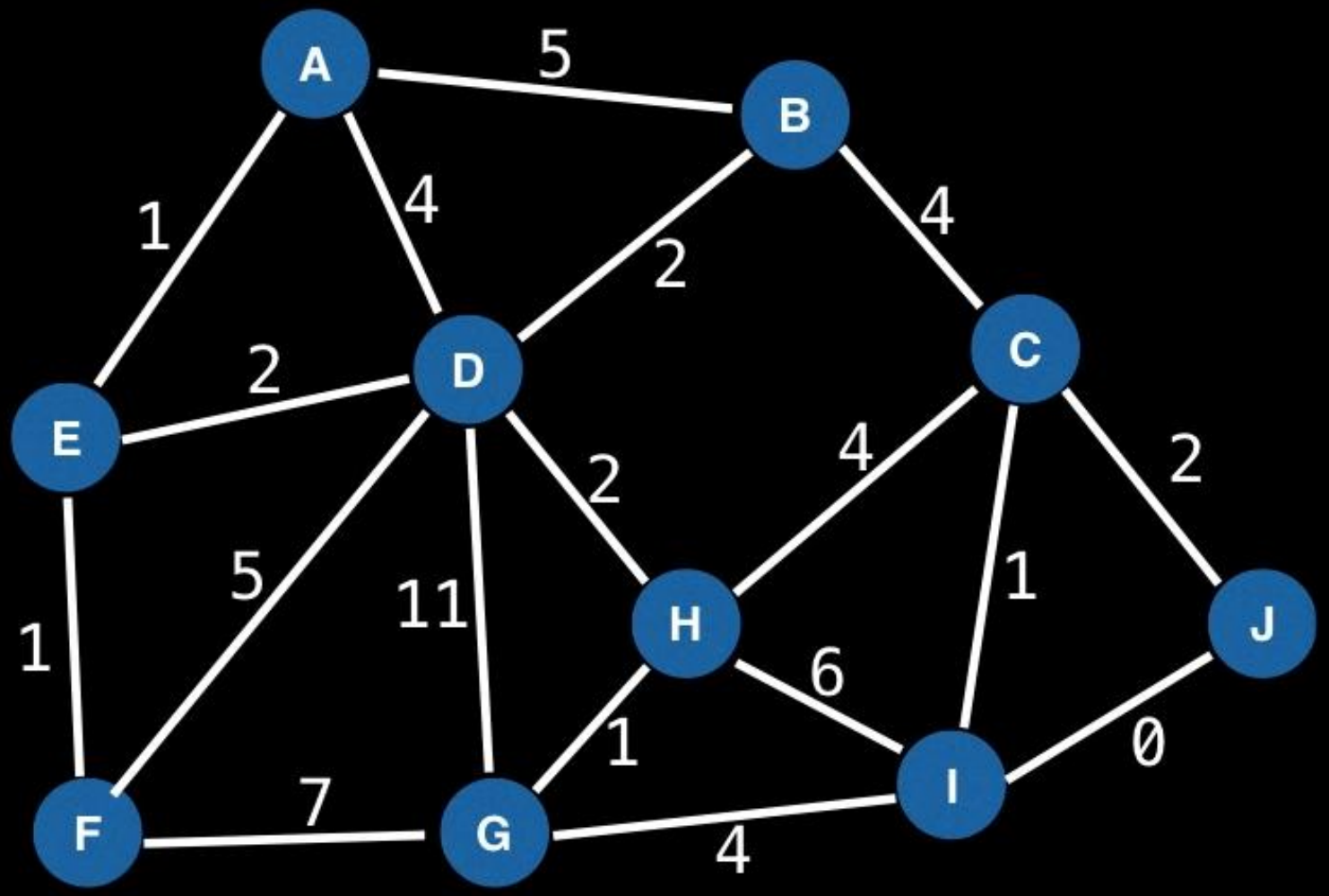
1. Sort the edges
2. Start with the cheapest

Kruskal’s algorithm

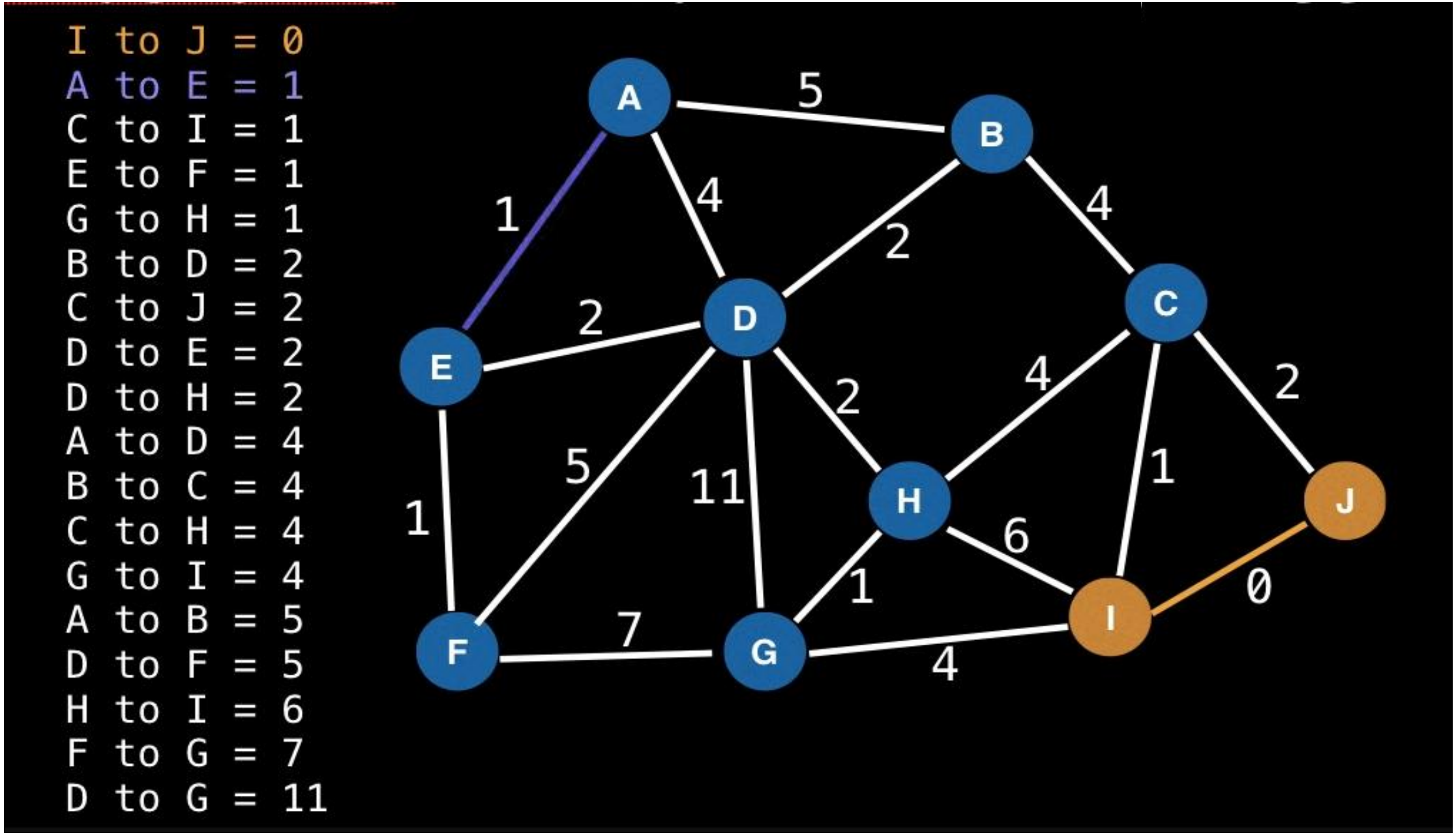
1. Sort the edges
2. Start with the cheapest
3. Keep building disjoint sets until they become one single set

https://www.youtube.com/watch?v=JZBQLXgSGfs

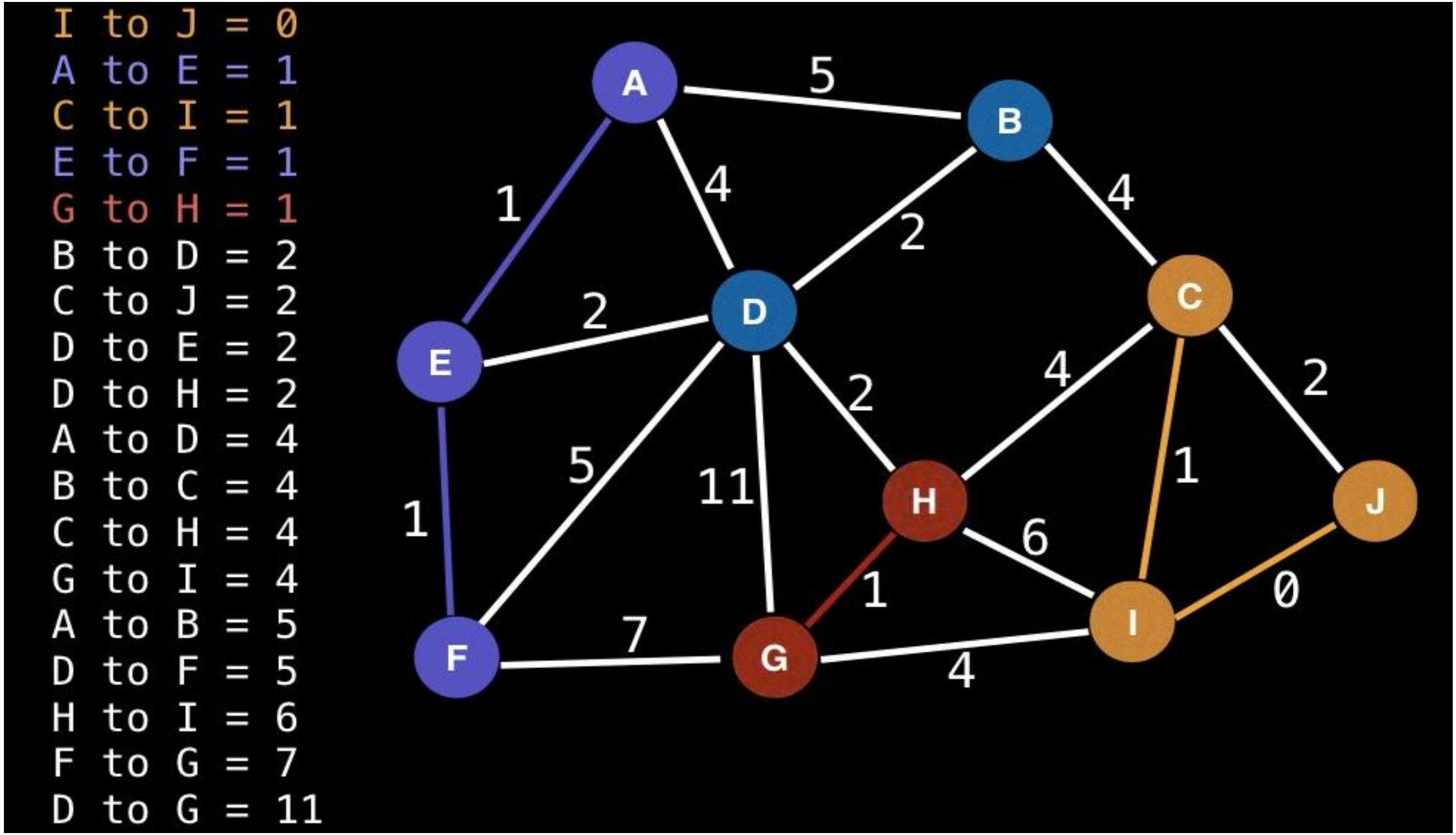
I to J = 0
A to E = 1
C to I = 1
E to F = 1
G to H = 1
B to D = 2
C to J = 2
D to E = 2
D to H = 2
A to D = 4
B to C = 4
C to H = 4
G to I = 4
A to B = 5
D to F = 5
H to I = 6
F to G = 7
D to G = 11



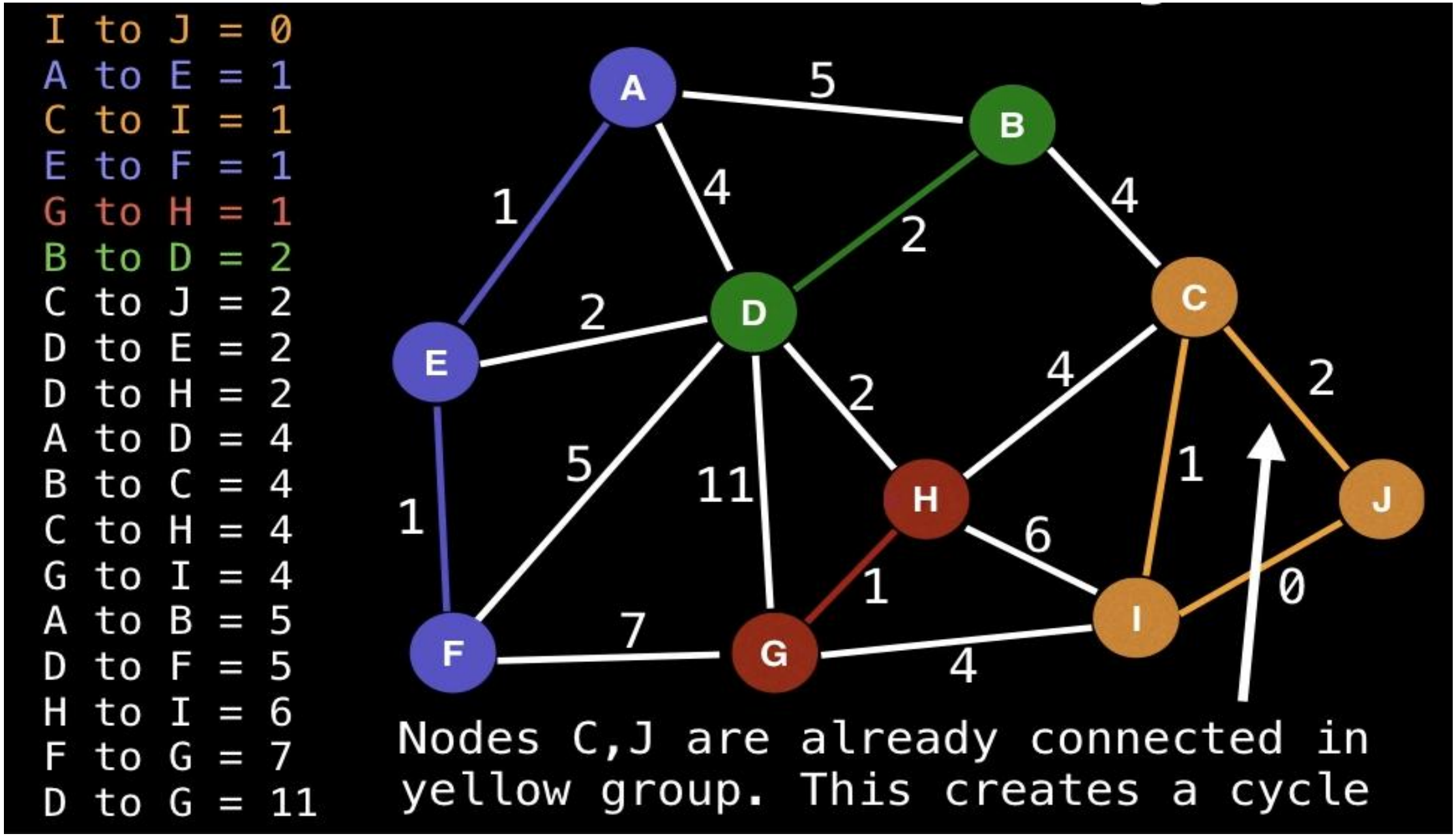
https://www.youtube.com/watch?v=JZBQLXgSGfs



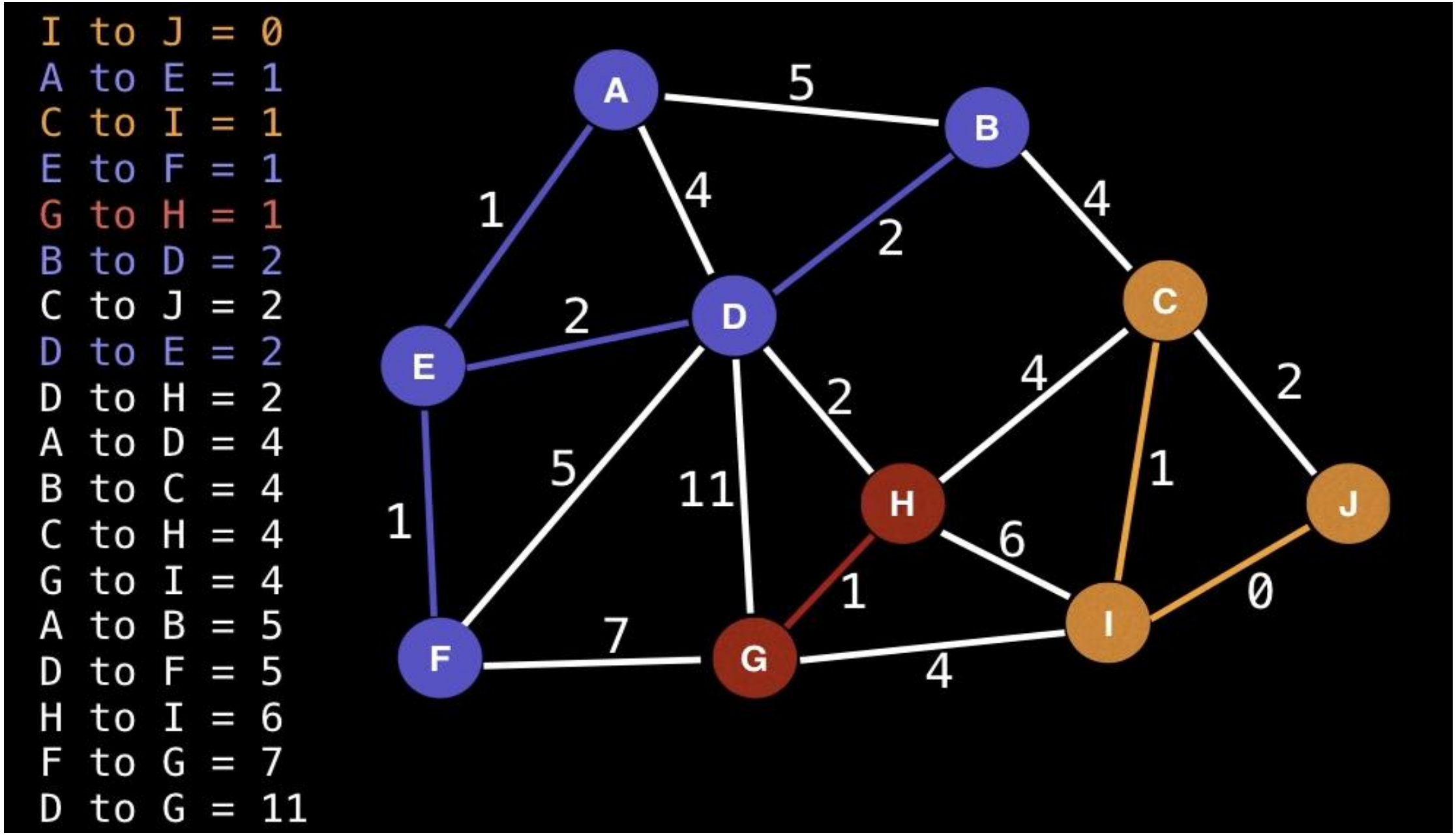
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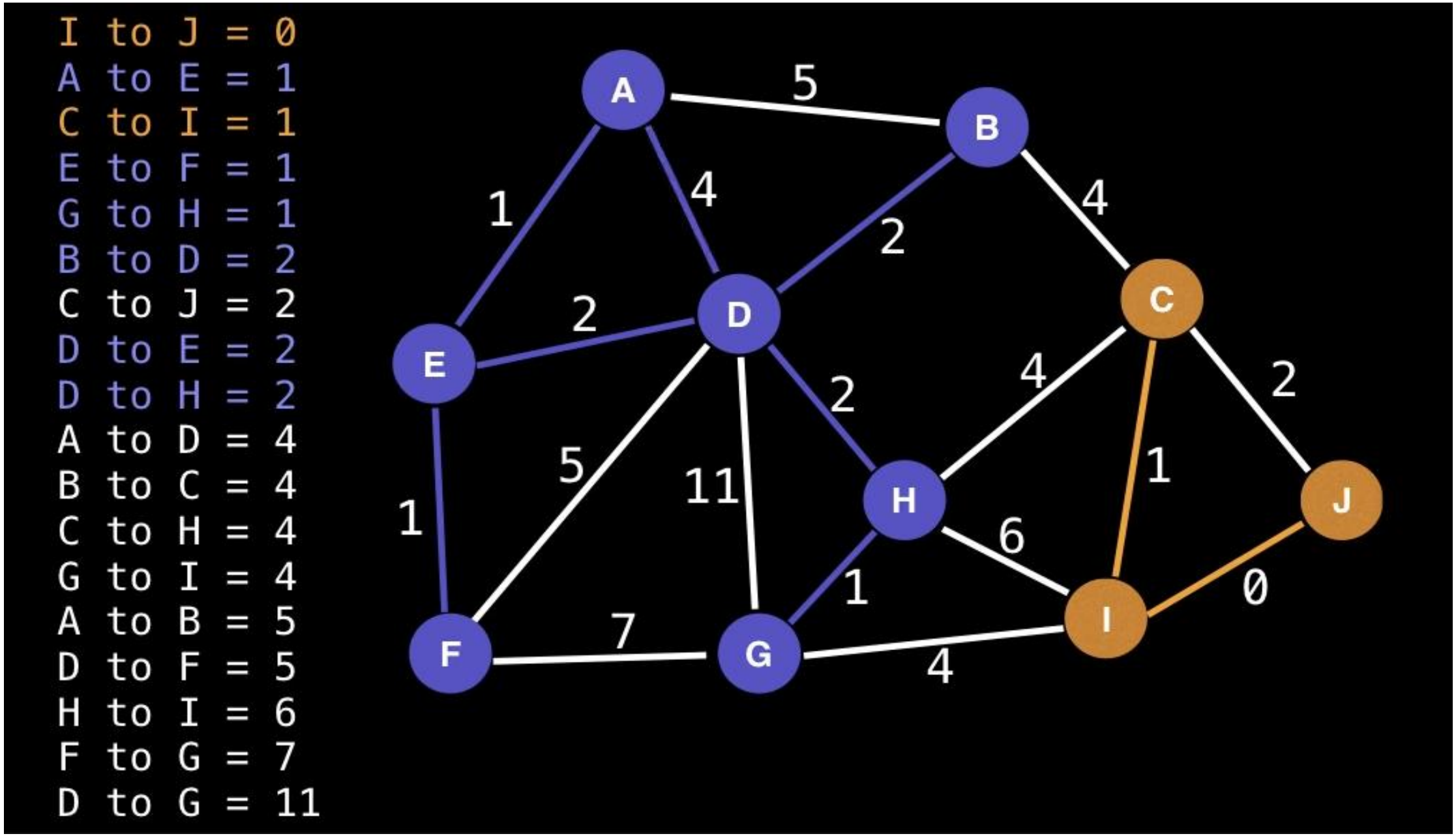
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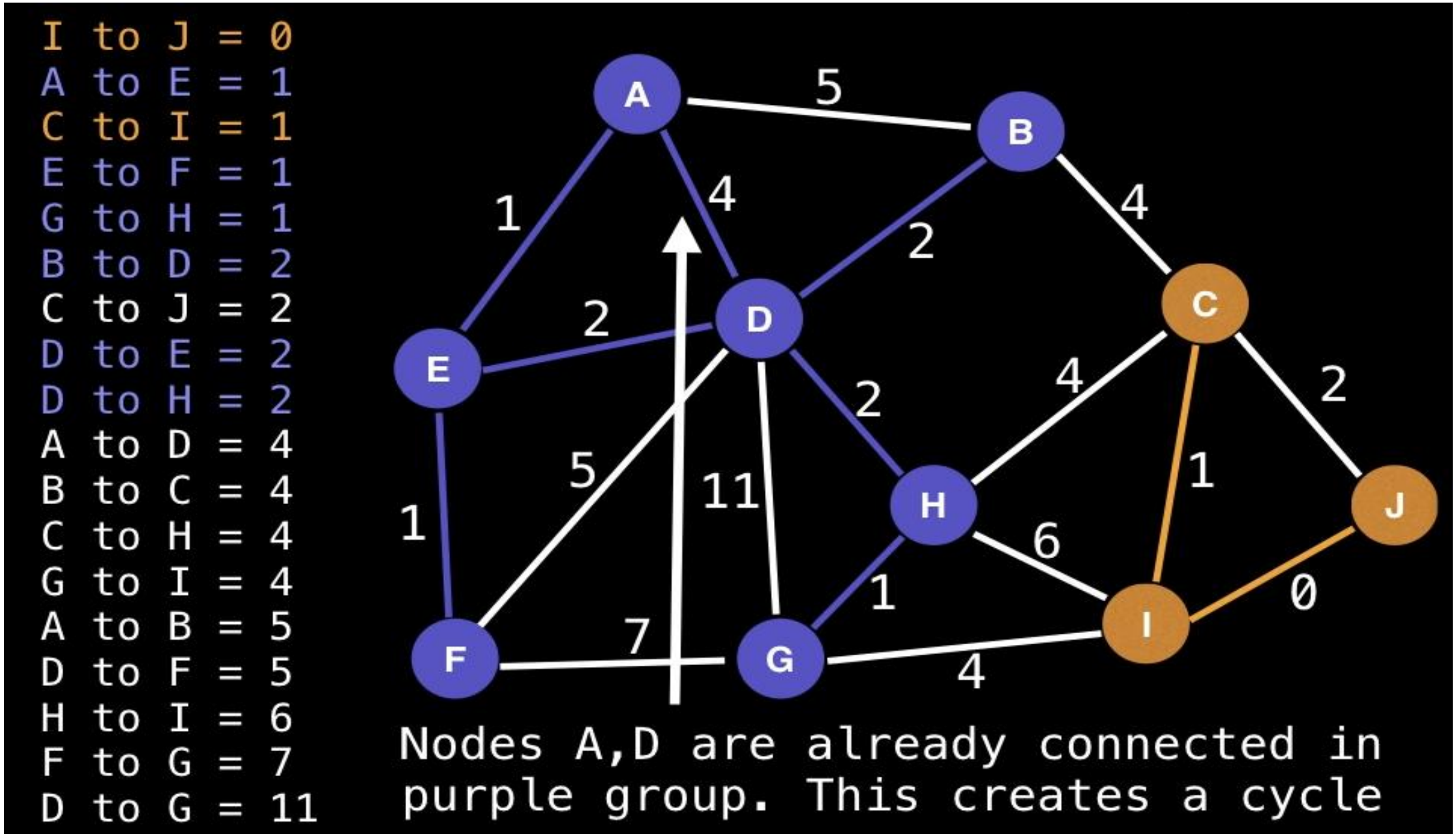
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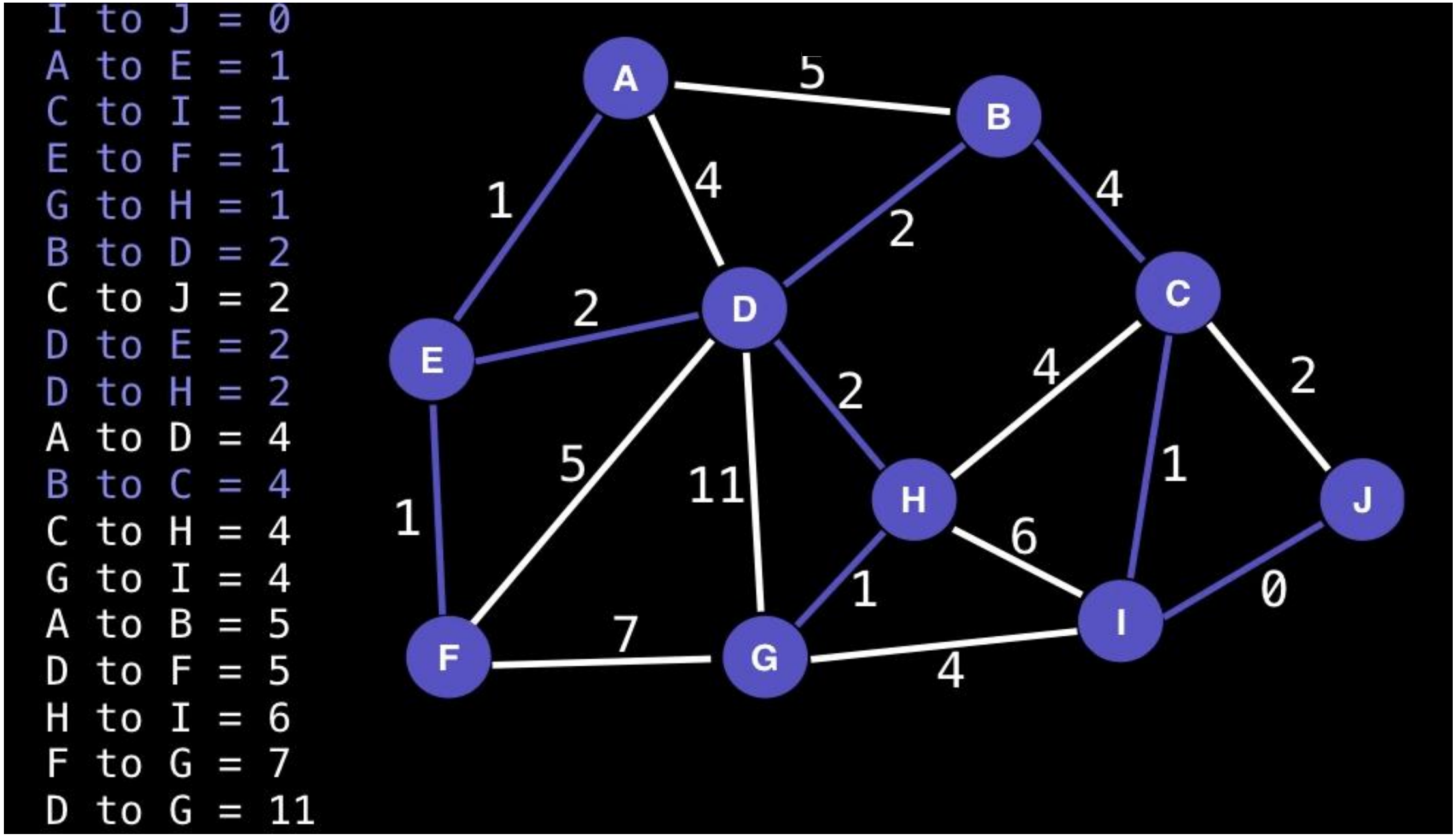
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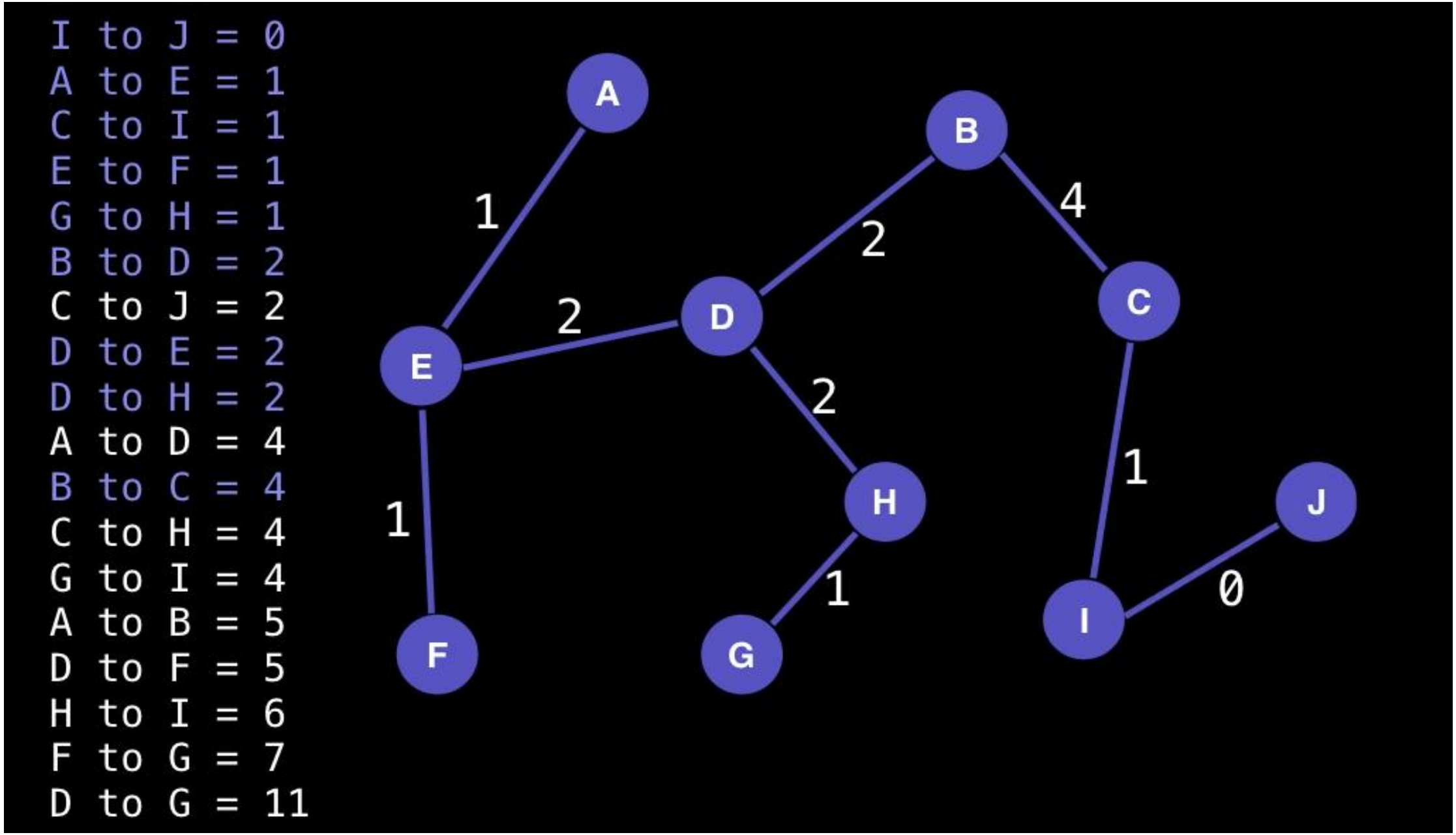
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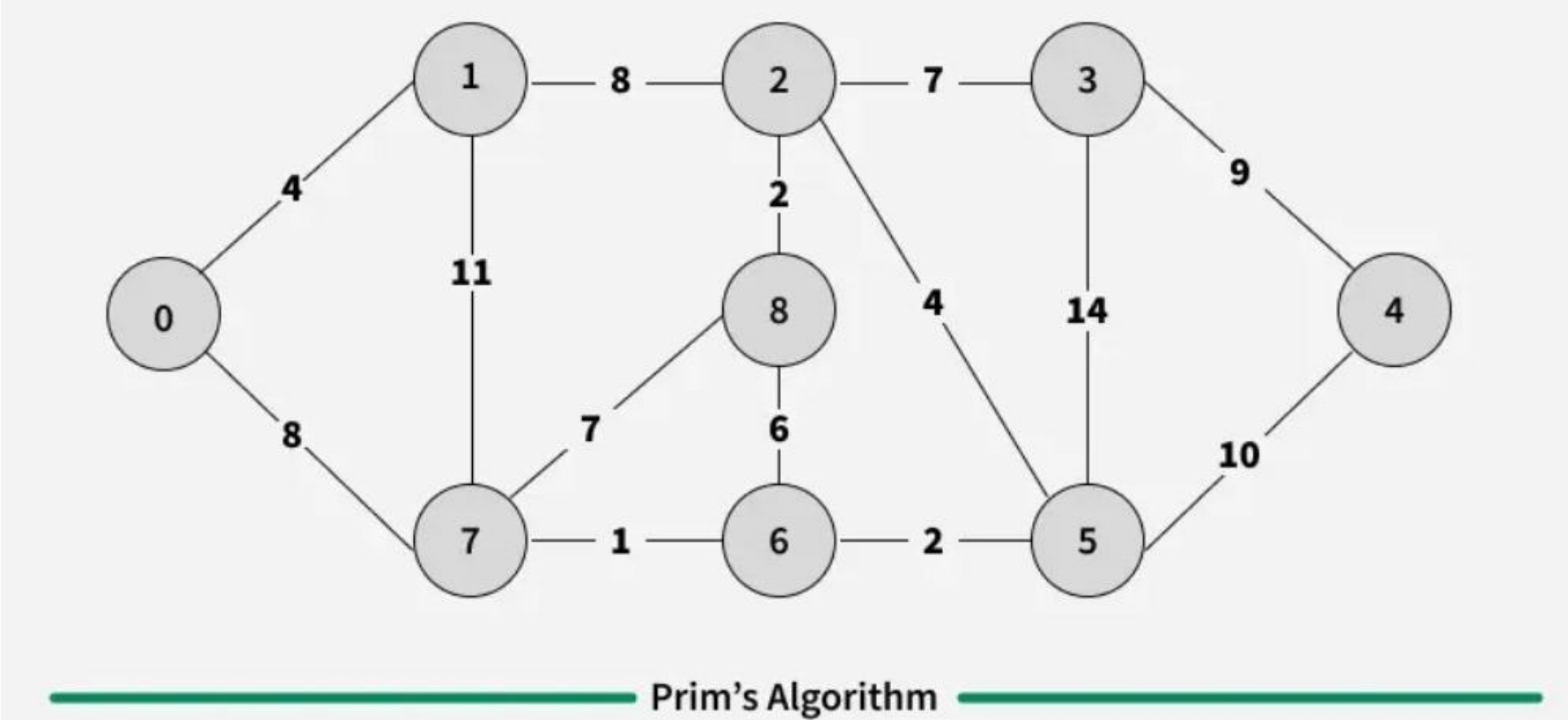


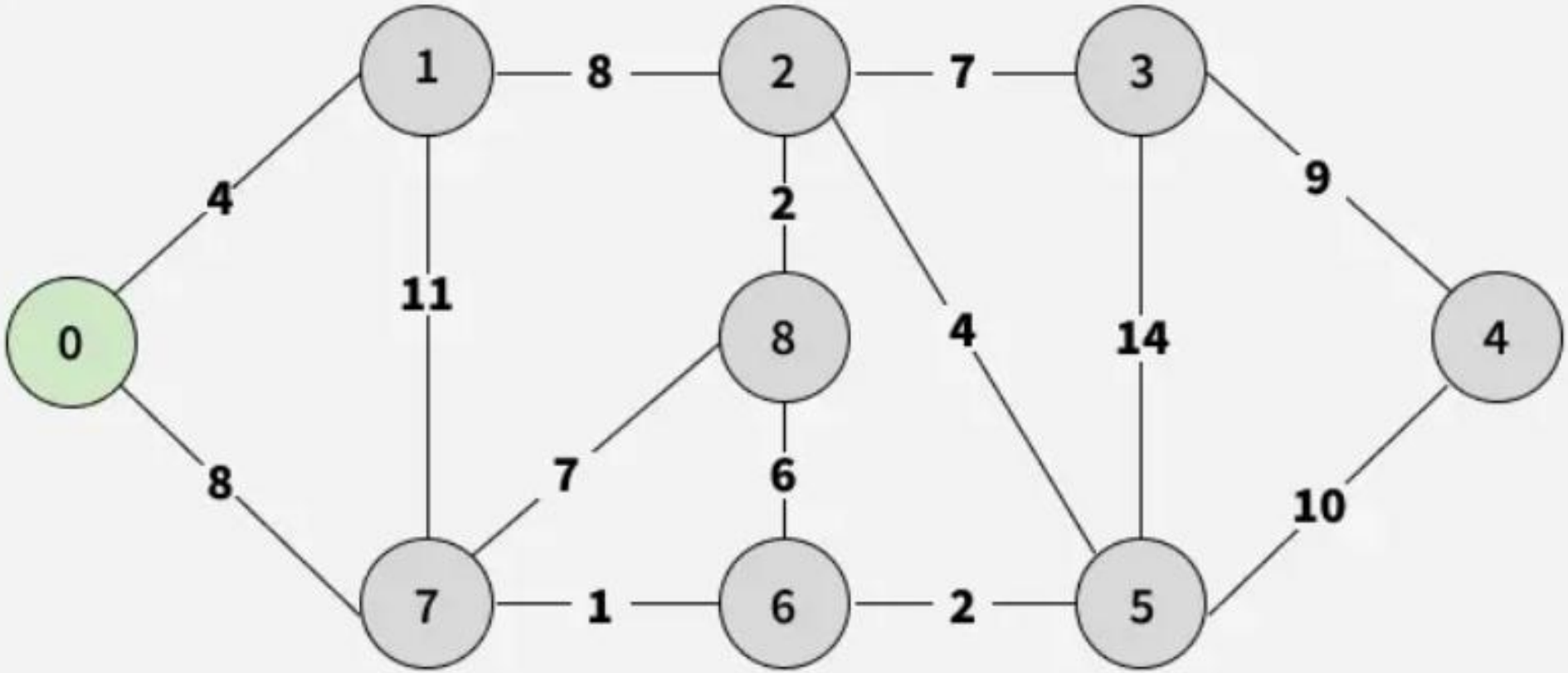
Prim’s algorithm

- https://www.youtube.com/watch?v=jsmMtJpPnhU&t=300s

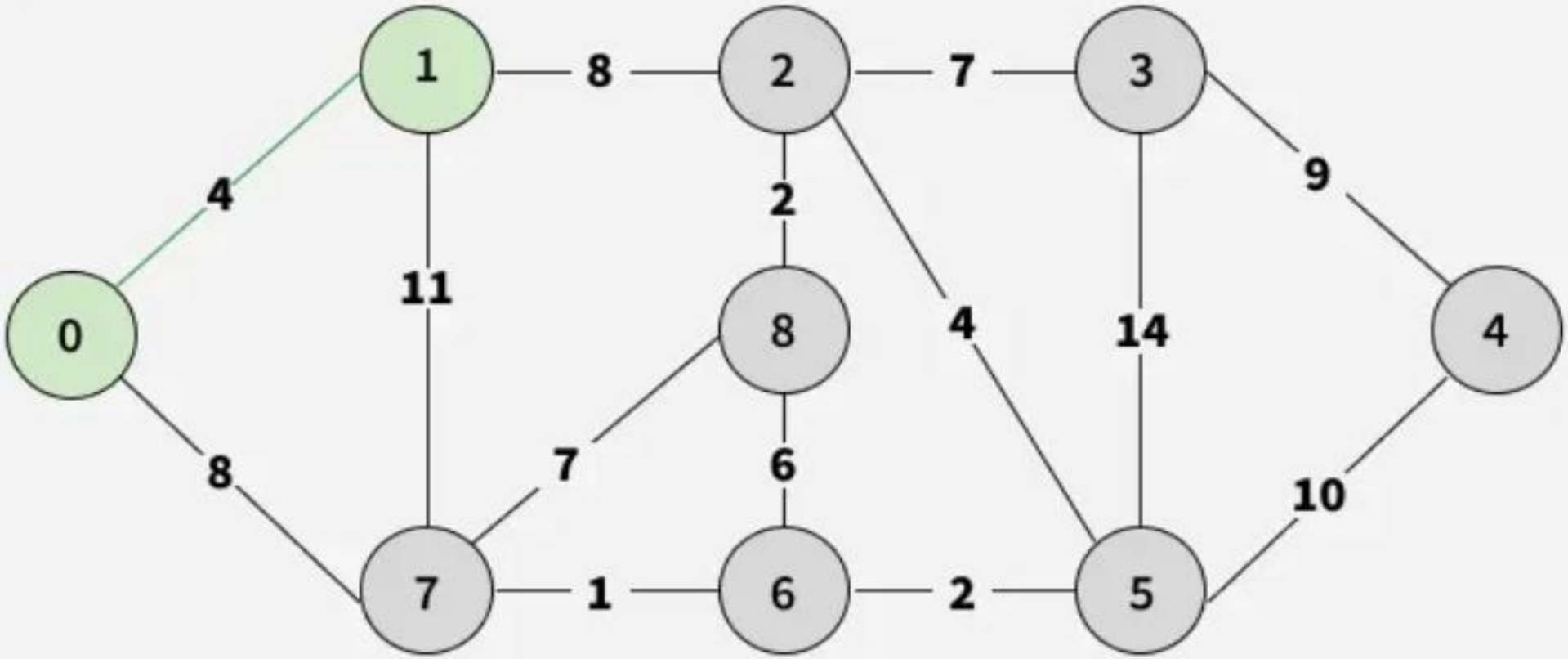
Prim’s algorithm – another example

https://www.geeksforgeeks.org/dsa/prims-minimum-spanning-tree-mst-greedy-algo-5/

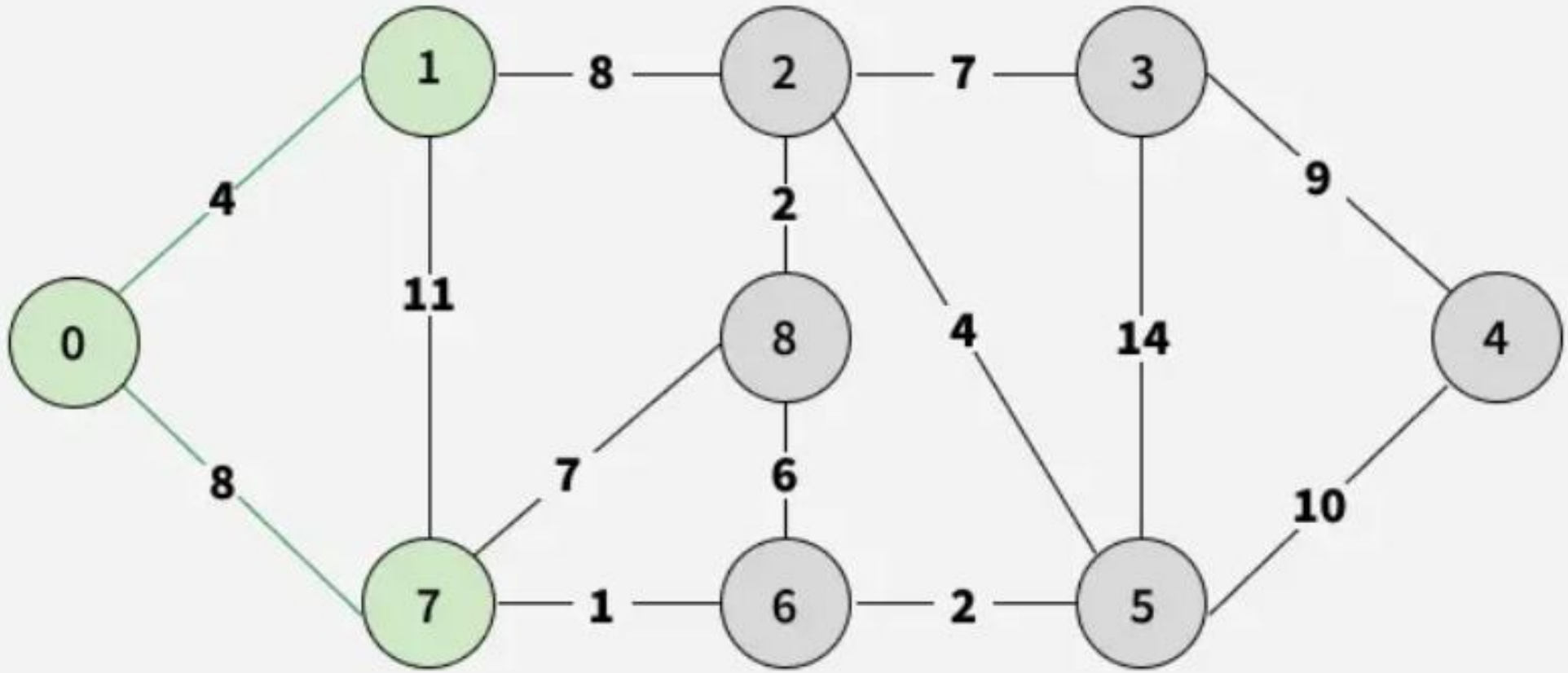




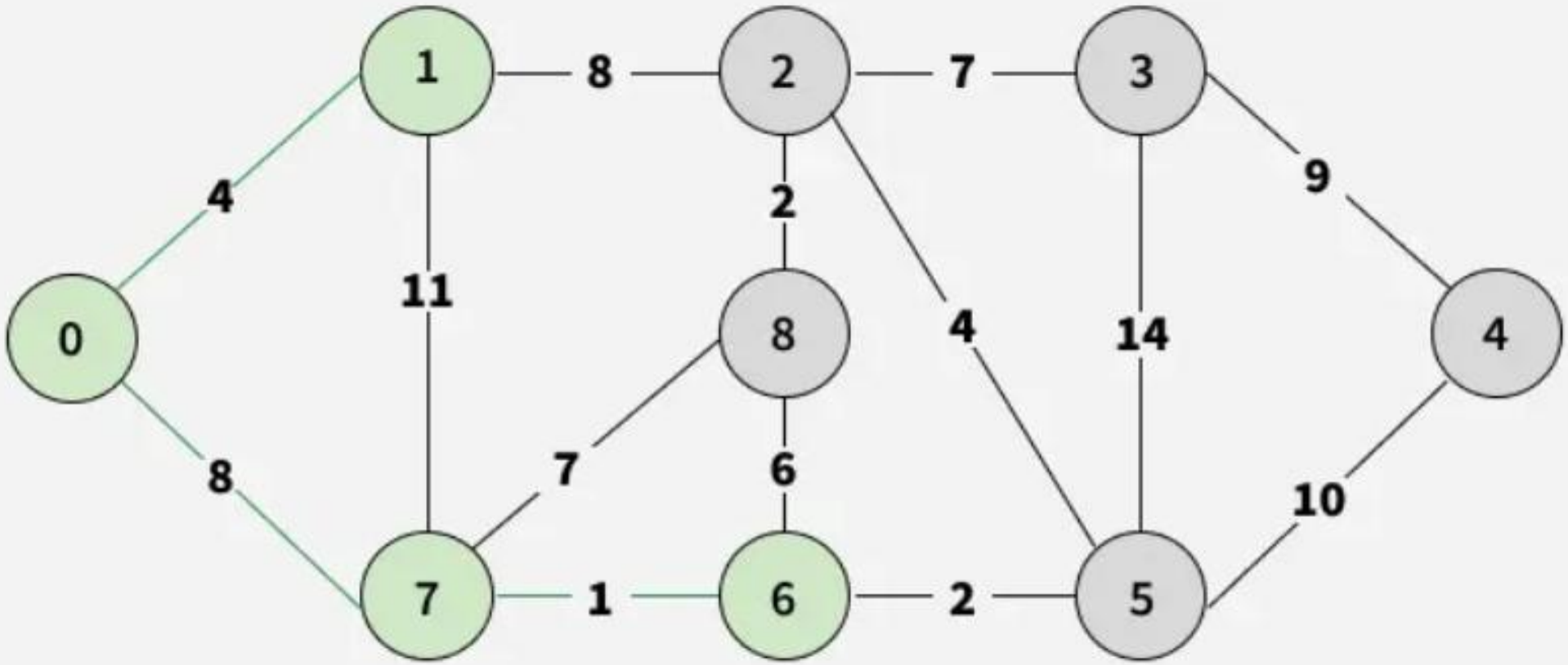
Prim's Algorithm



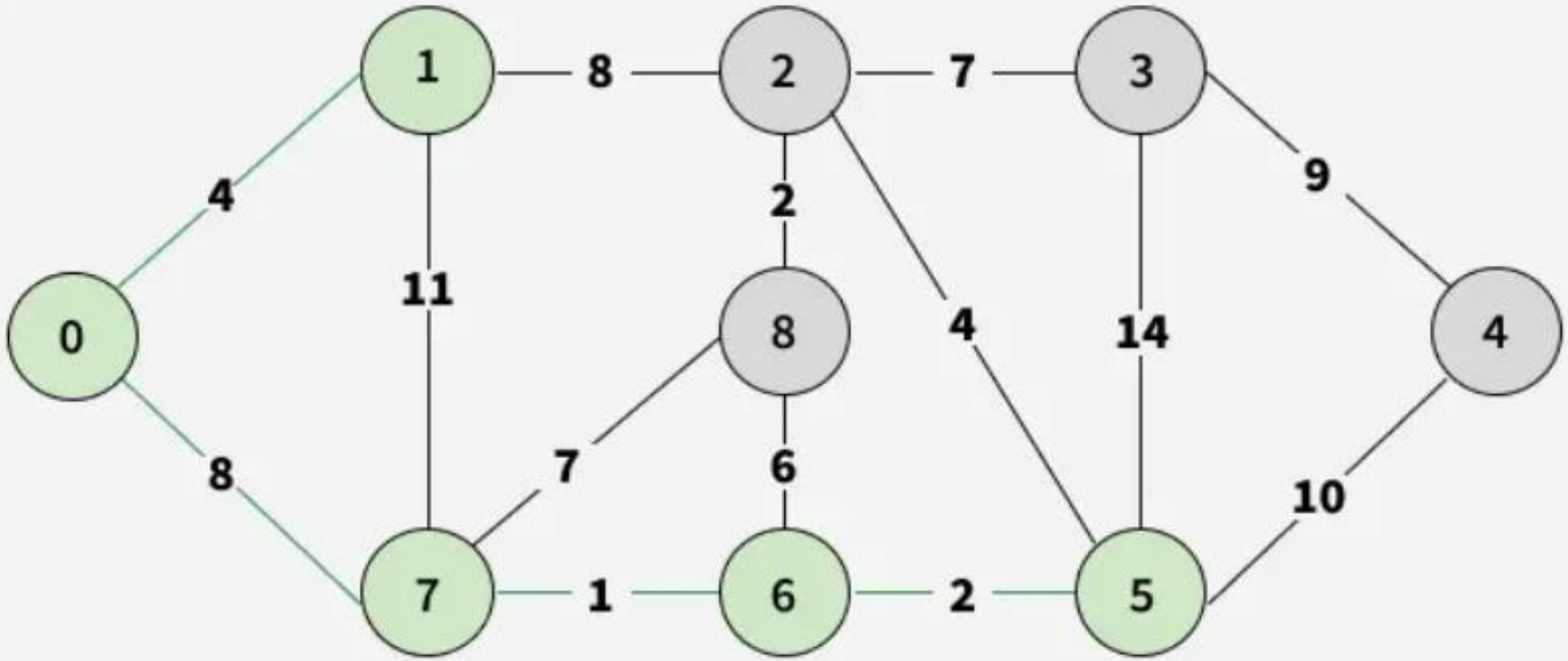
Prim's Algorithm



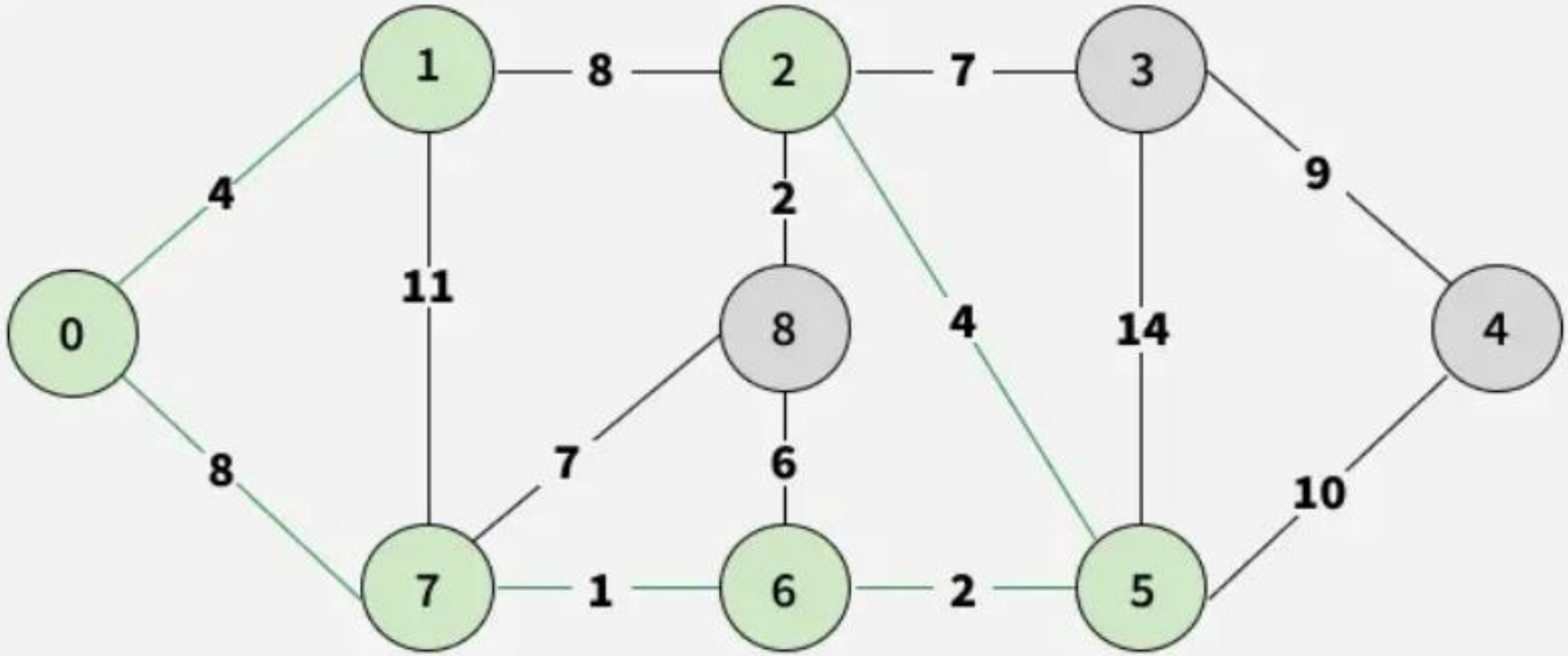
Prim's Algorithm



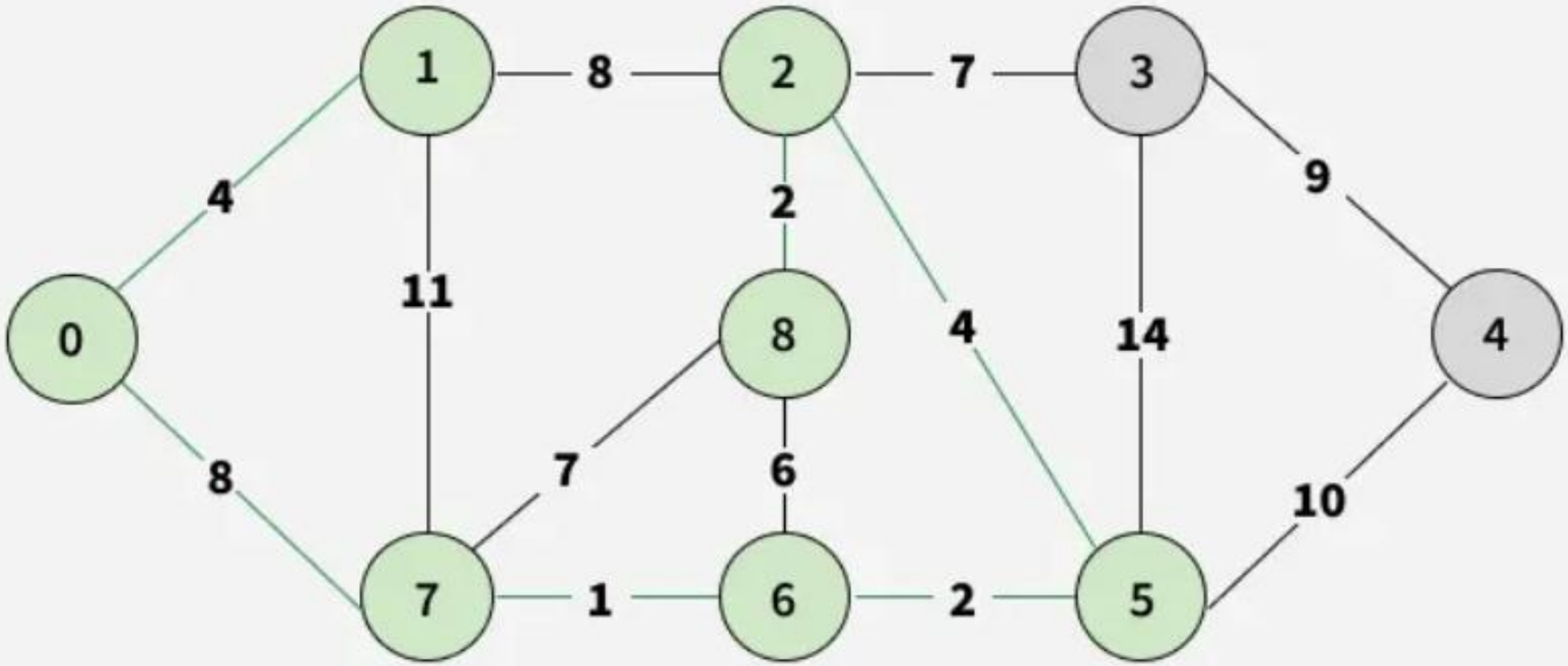
Prim's Algorithm



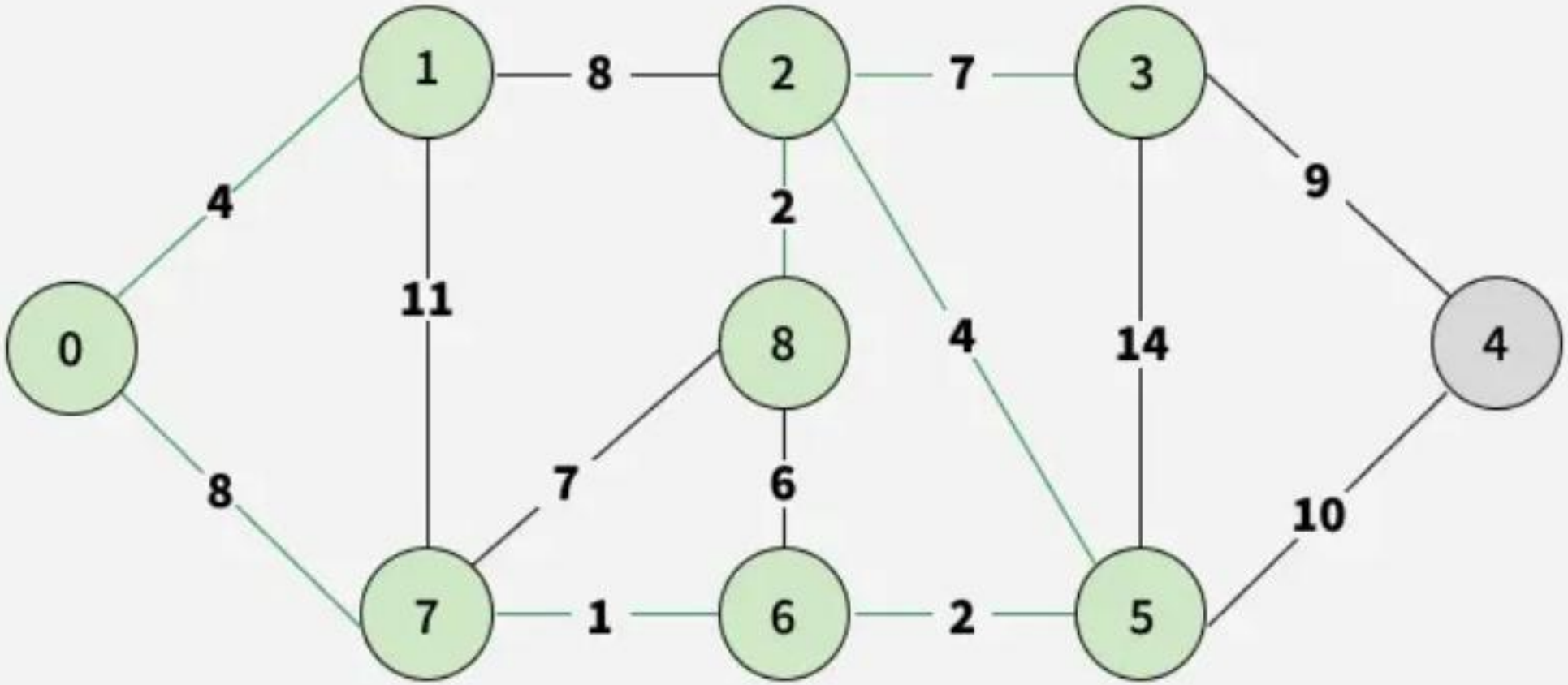
Prim's Algorithm



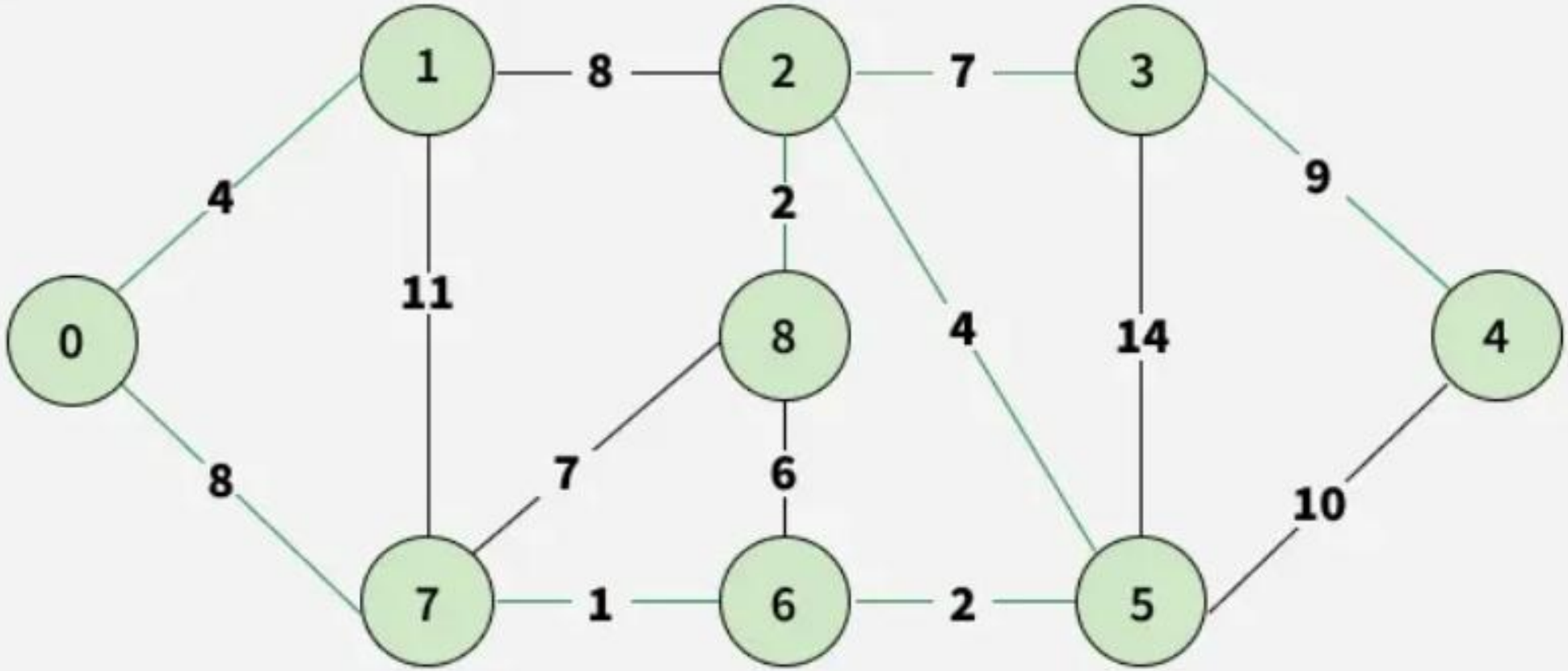
Prim's Algorithm



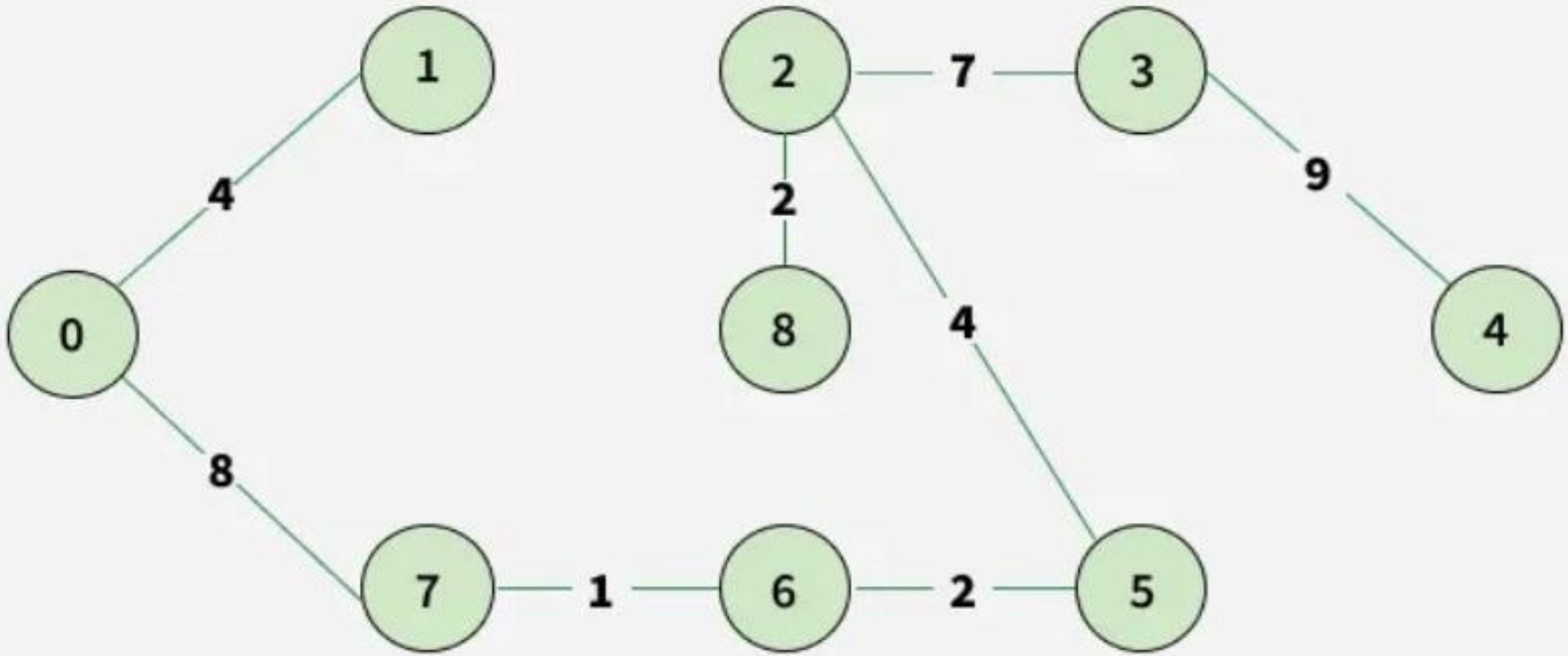
Prim's Algorithm



Prim's Algorithm



Prim's Algorithm



Prim's Algorithm



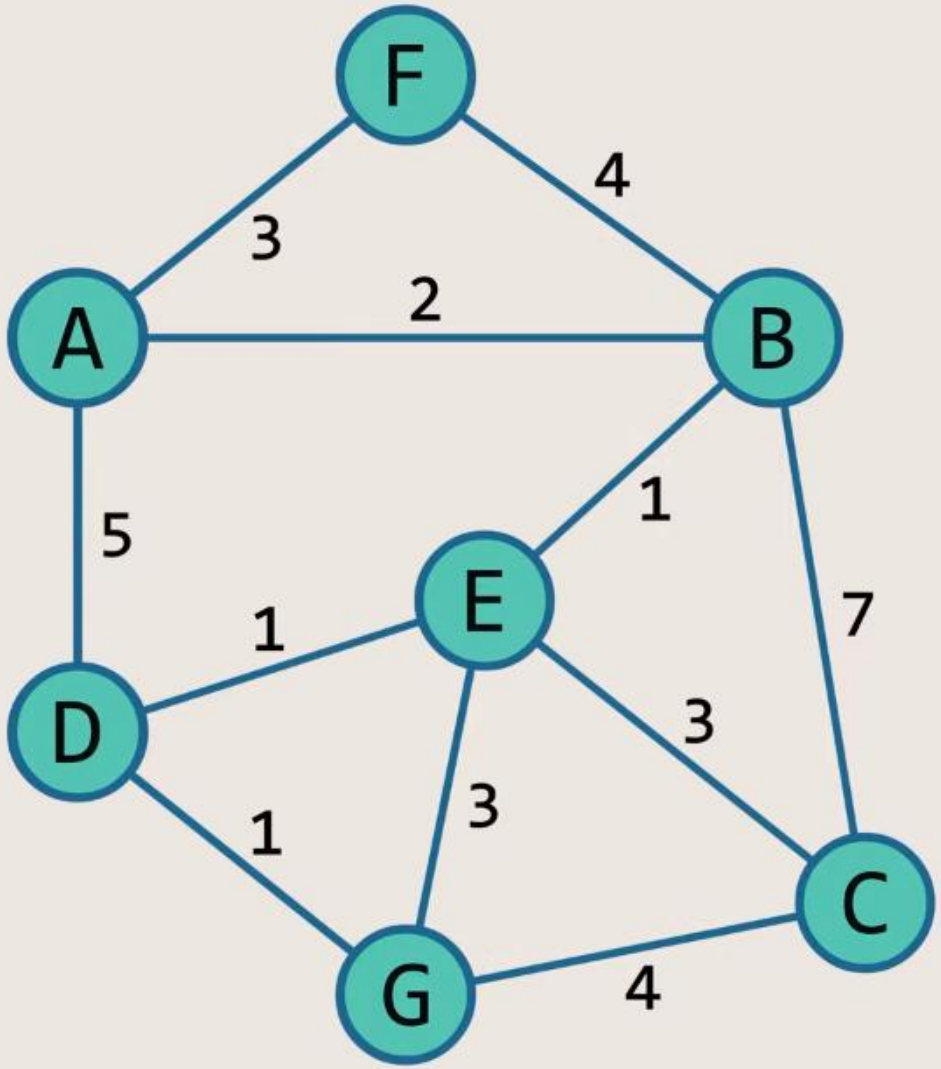
Shortest path problem

Single Source Shortest Path (SSSP)

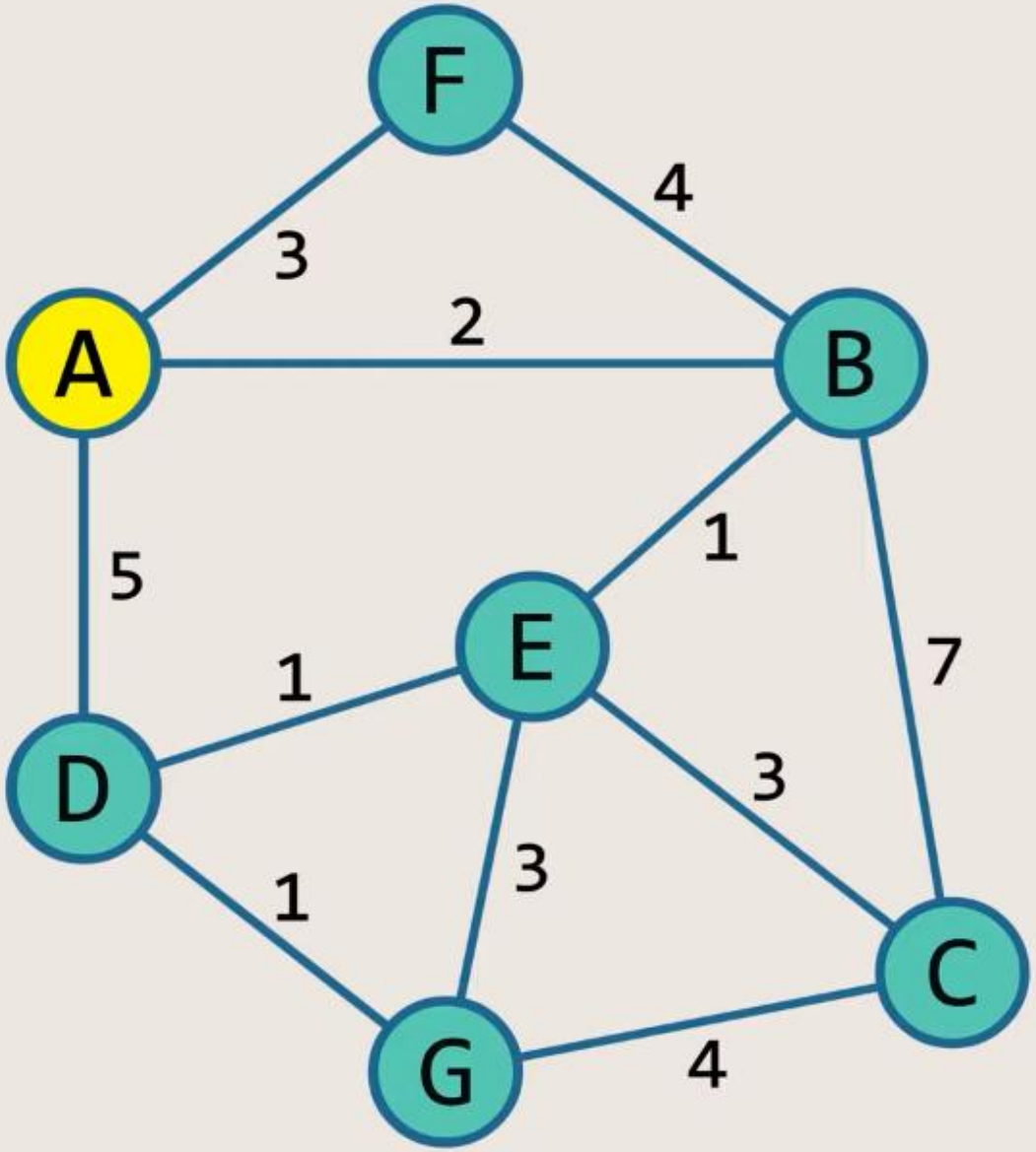
Dijkstra’s algorithm

https://www.youtube.com/watch?v=bZkzH5x0SKU

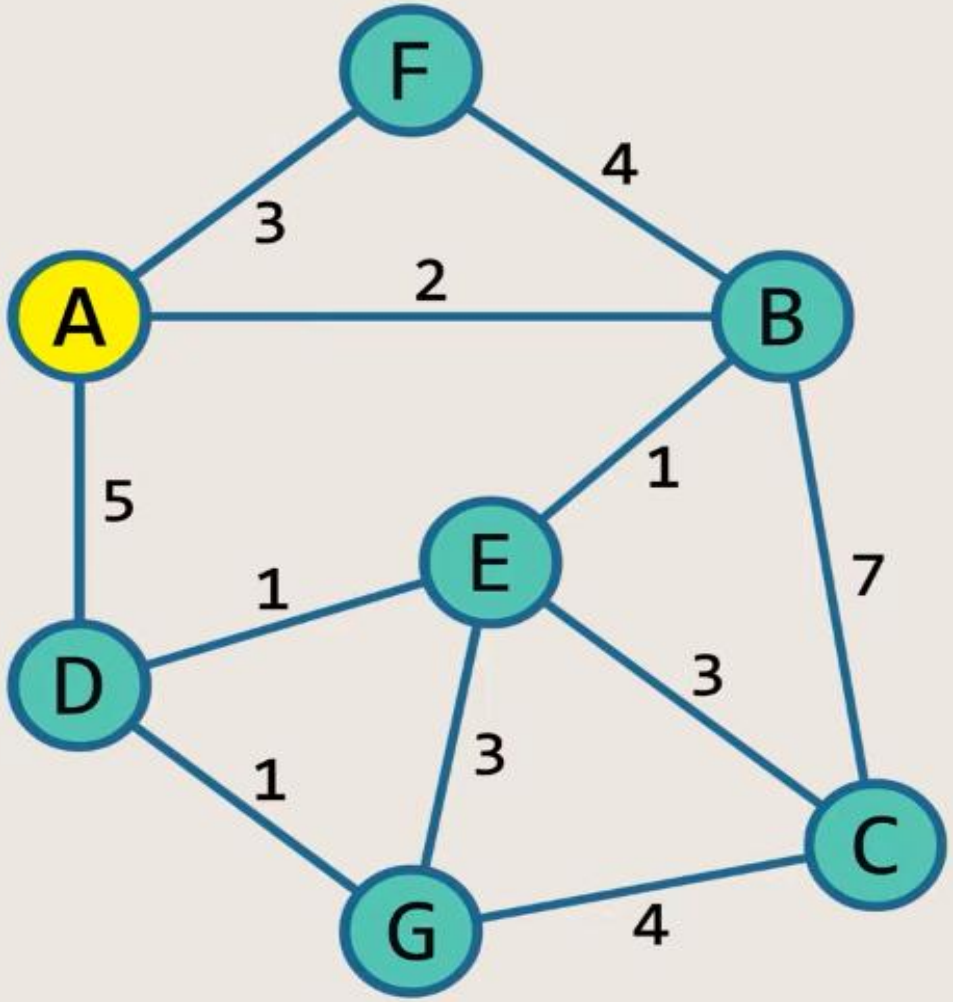
https://www.youtube.com/watch?v=CmIQ29cUGiE



Node	Cost	Previous
A		
B		
C		
D		
E		
F		
G		

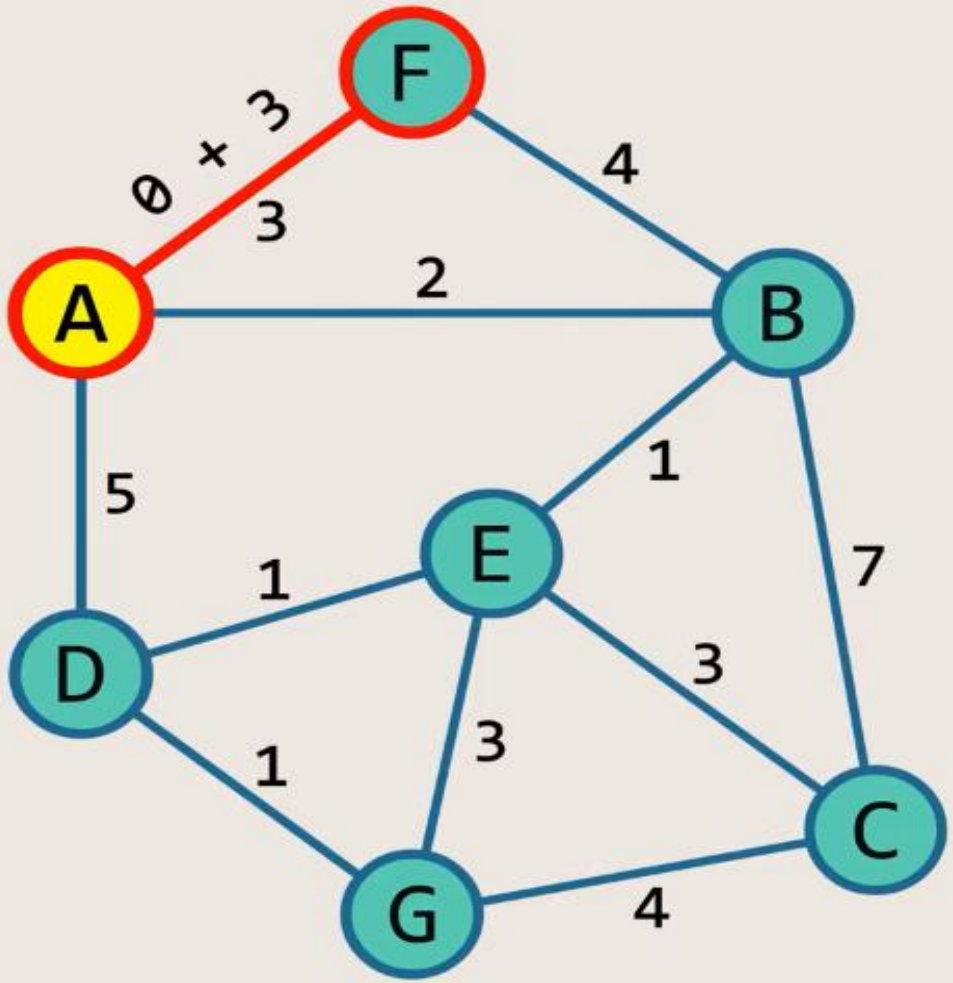


Node	Cost	Previous
A	0	
B	∞	
C	∞	
D	∞	
E	∞	
F	∞	
G	∞	

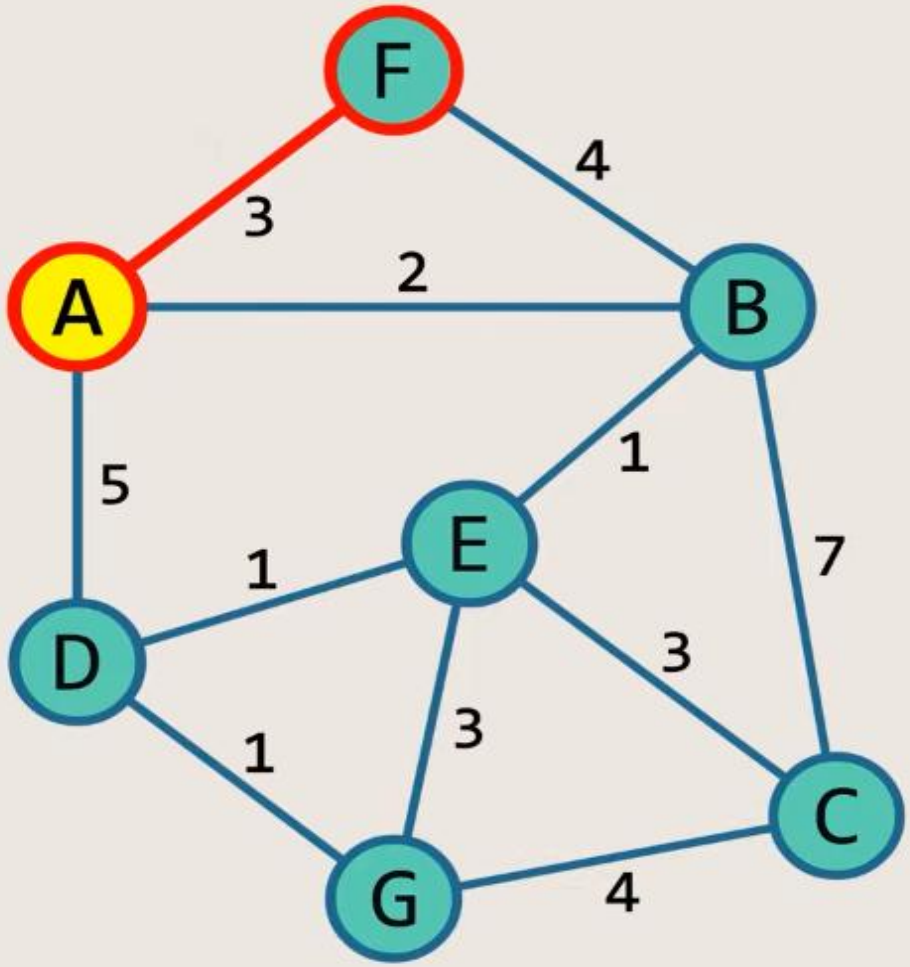


Node	Cost	Previous
A	0	
B	∞	
C	∞	
D	∞	
E	∞	
F	∞	
G	∞	

A,0

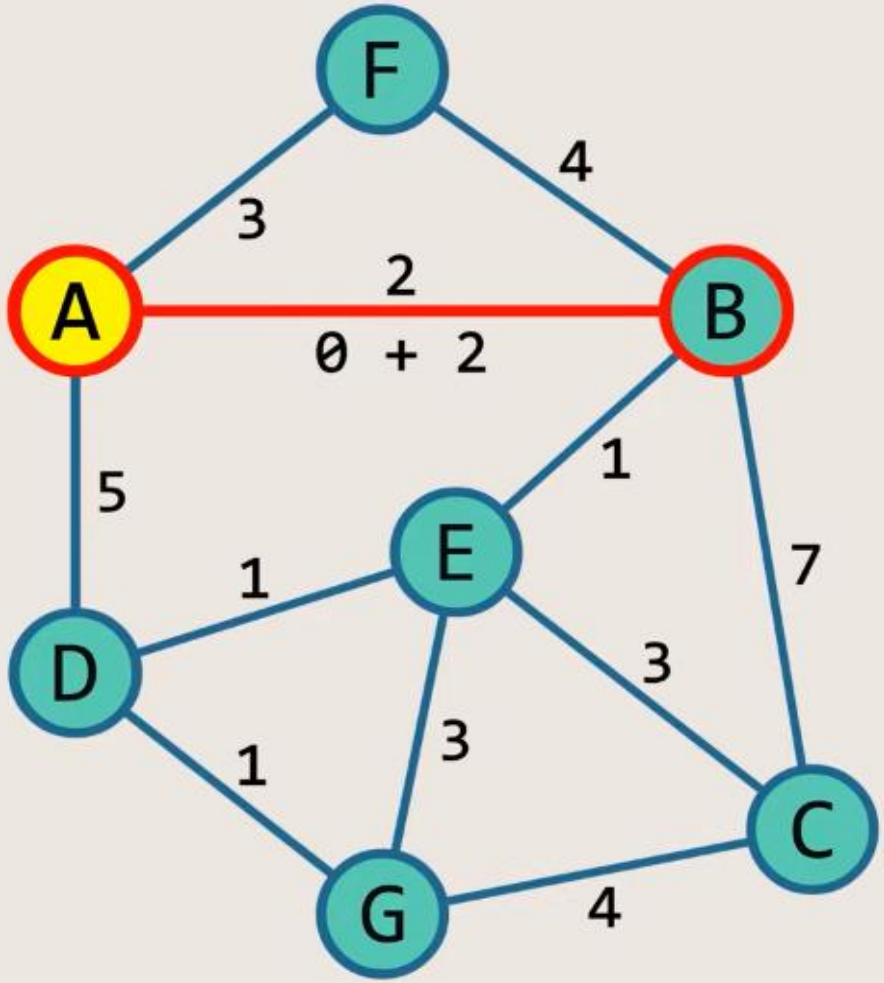


Node	Cost	Previous
A	0	
B	∞	
C	∞	
D	∞	
E	∞	
F	∞	
G	∞	



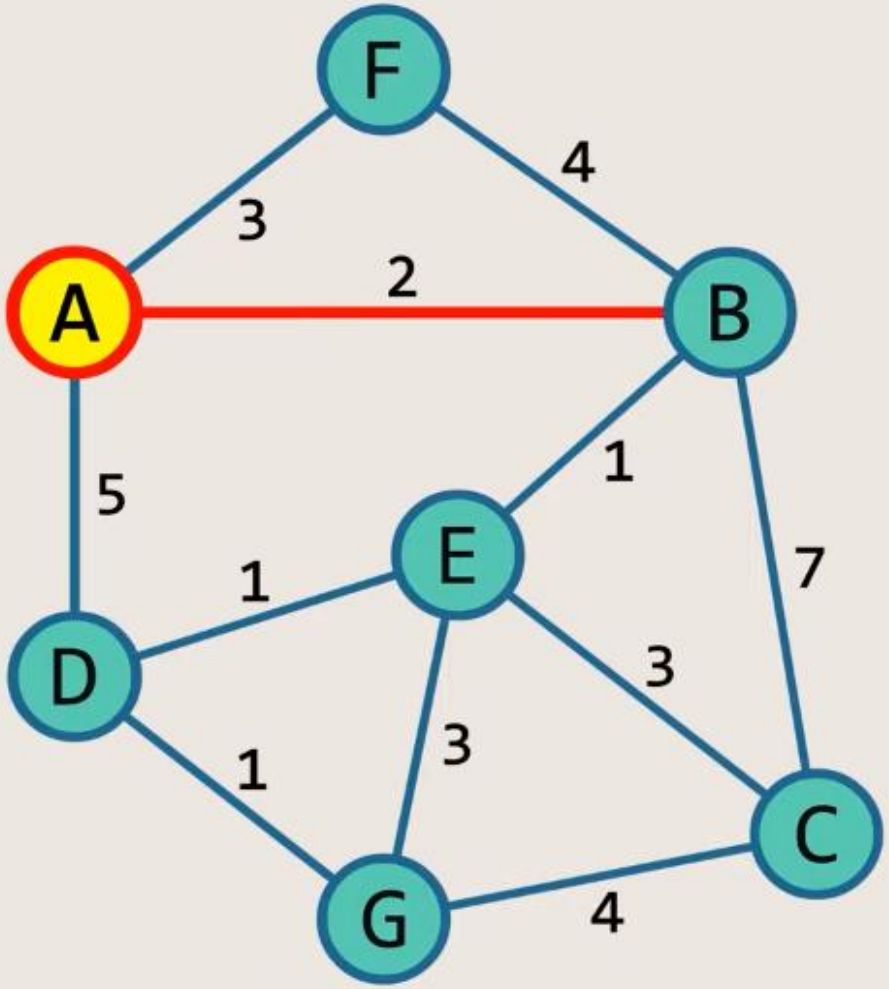
Node	Cost	Previous
A	0	
B	∞	
C	∞	
D	∞	
E	∞	
F	3	A
G	∞	

F,3



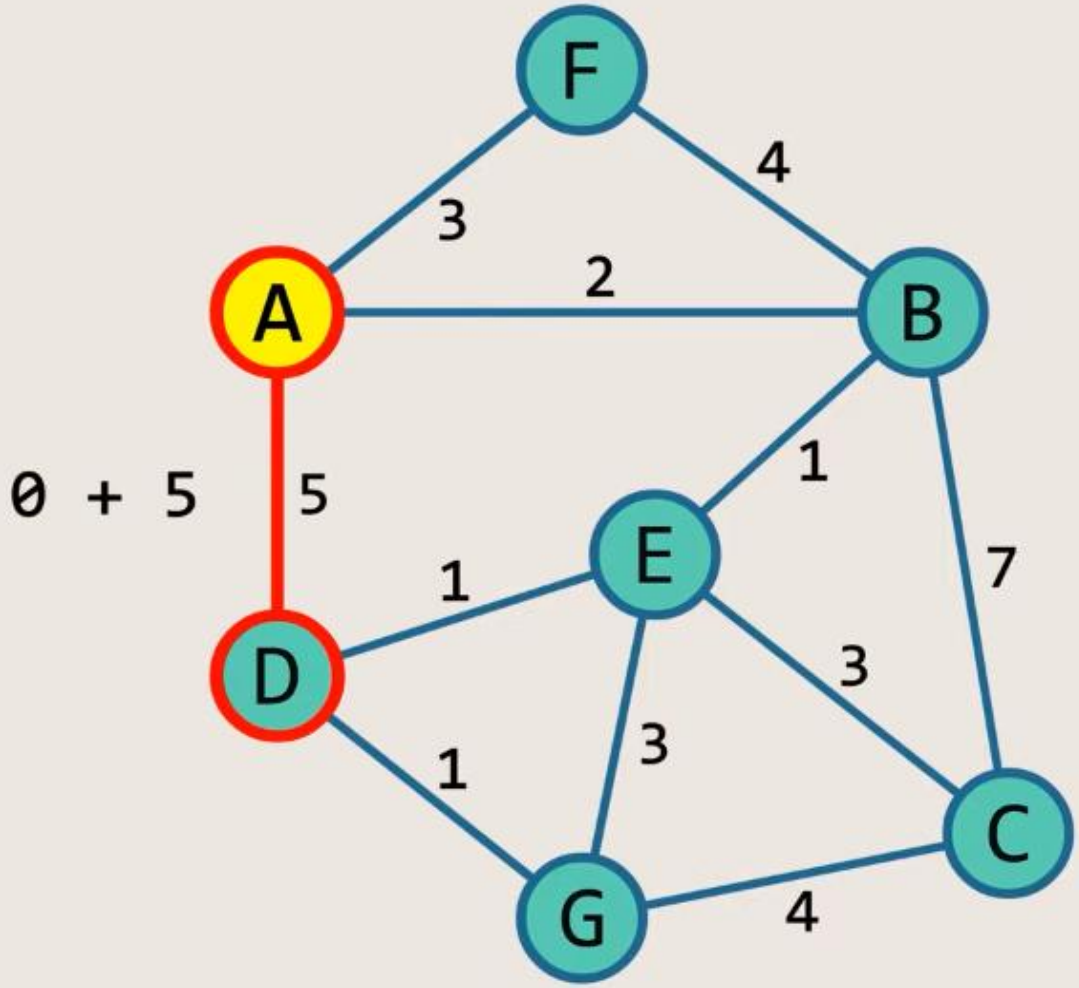
Node	Cost	Previous
A	0	
B	∞	
C	∞	
D	∞	
E	∞	
F	3	A
G	∞	

F,3



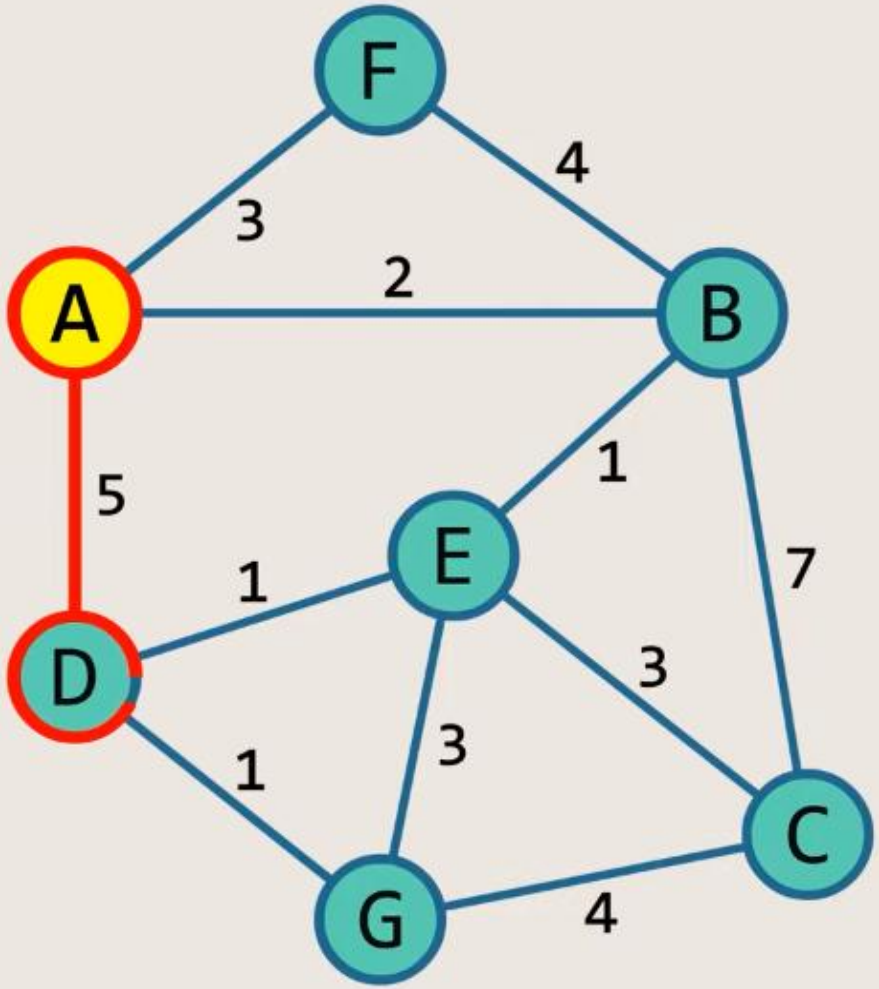
Node	Cost	Previous
A	0	
B	2	A
C	∞	
D	∞	
E	∞	
F	3	A
G	∞	

F,3 B,2



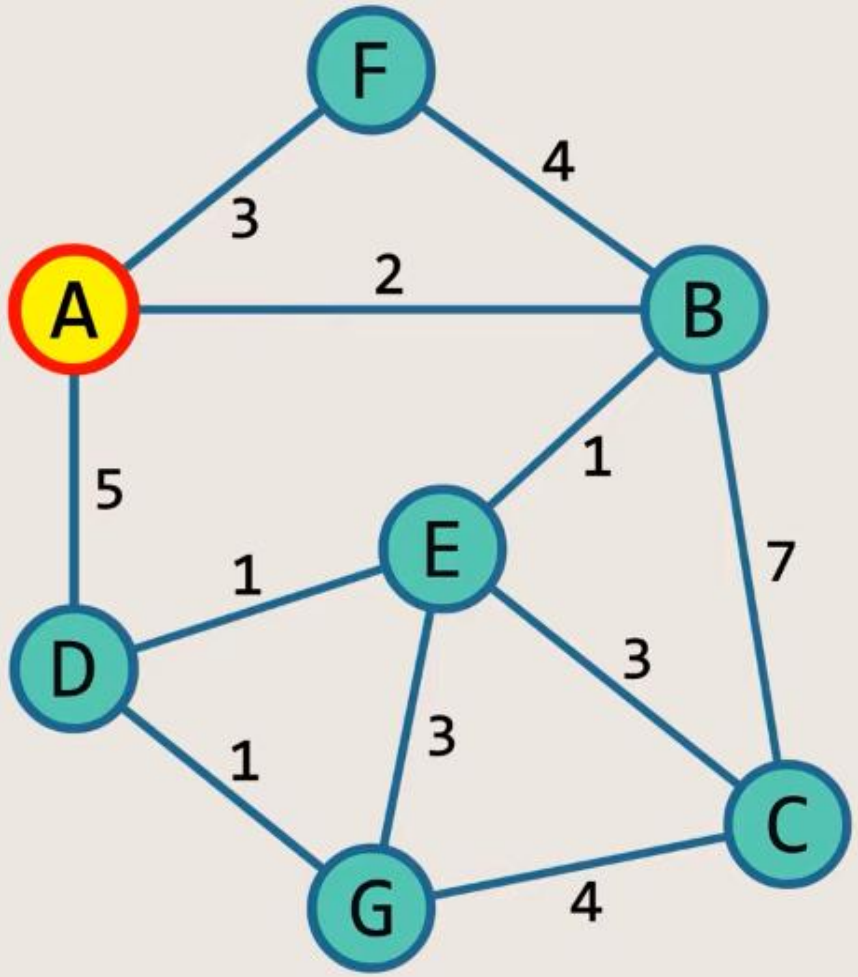
Node	Cost	Previous
A	0	
B	2	A
C	∞	
D	∞	
E	∞	
F	3	A
G	∞	

F,3 B,2



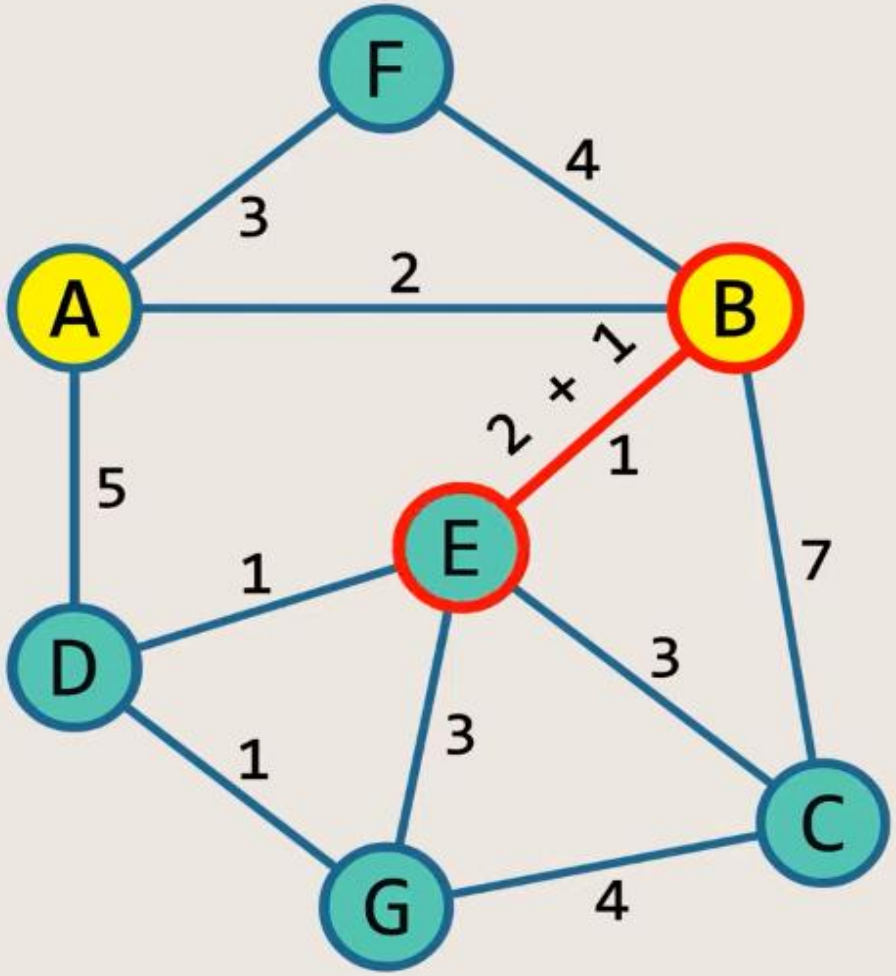
Node	Cost	Previous
A	0	
B	2	A
C	∞	
D	5	A
E	∞	
F	3	A
G	∞	





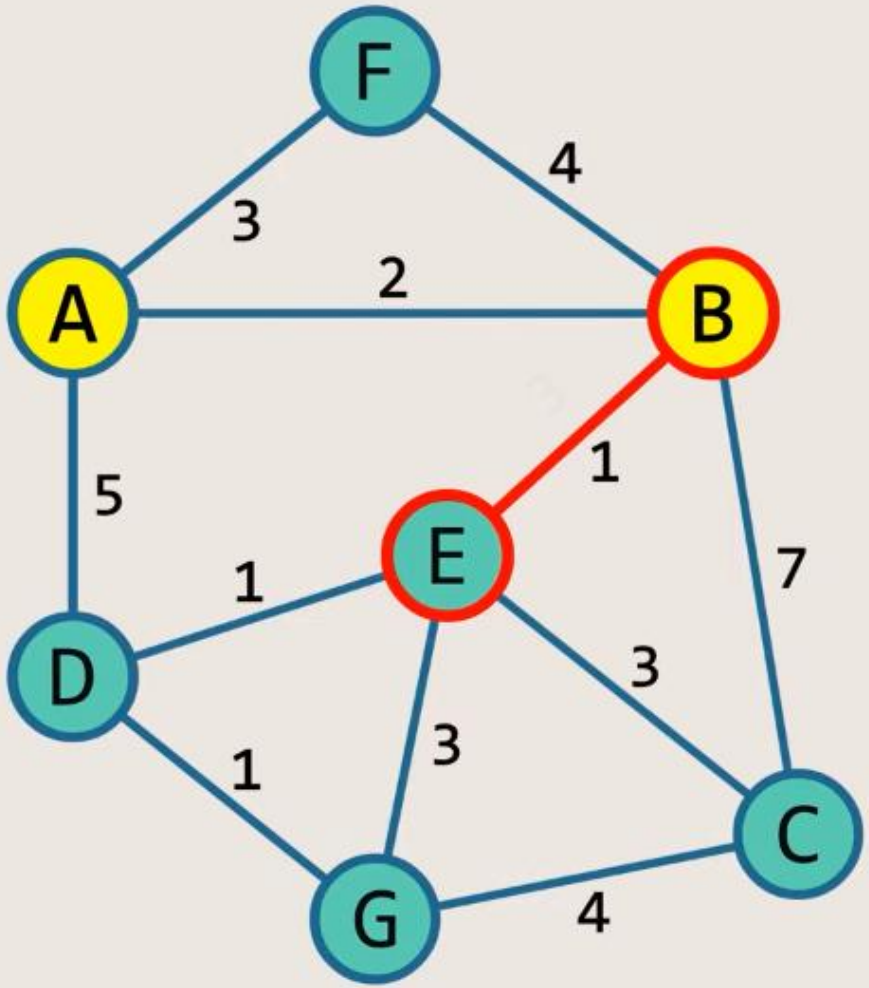
Node	Cost	Previous
A	0	
B	2	A
C	∞	
D	5	A
E	∞	
F	3	A
G	∞	

B,2 | F,3 | D,5



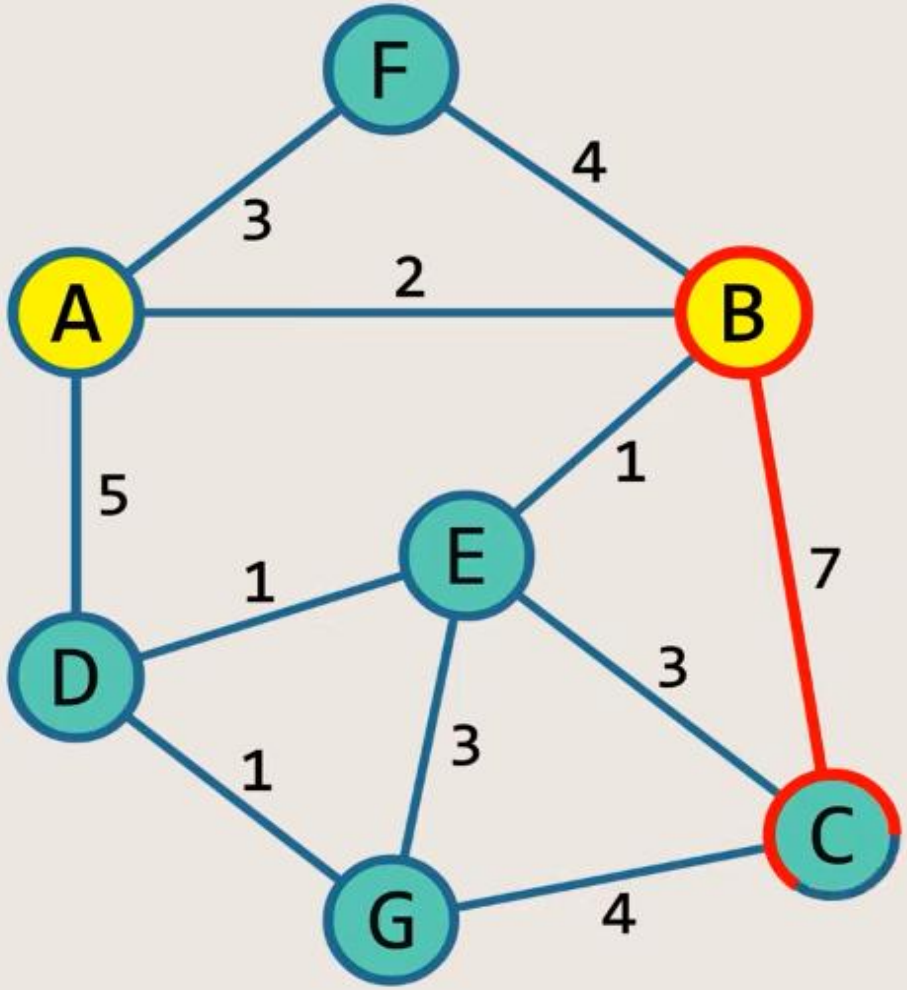
Node	Cost	Previous
A	0	
B	2	A
C	∞	
D	5	A
E	∞	
F	3	A
G	∞	

F,3 D,5



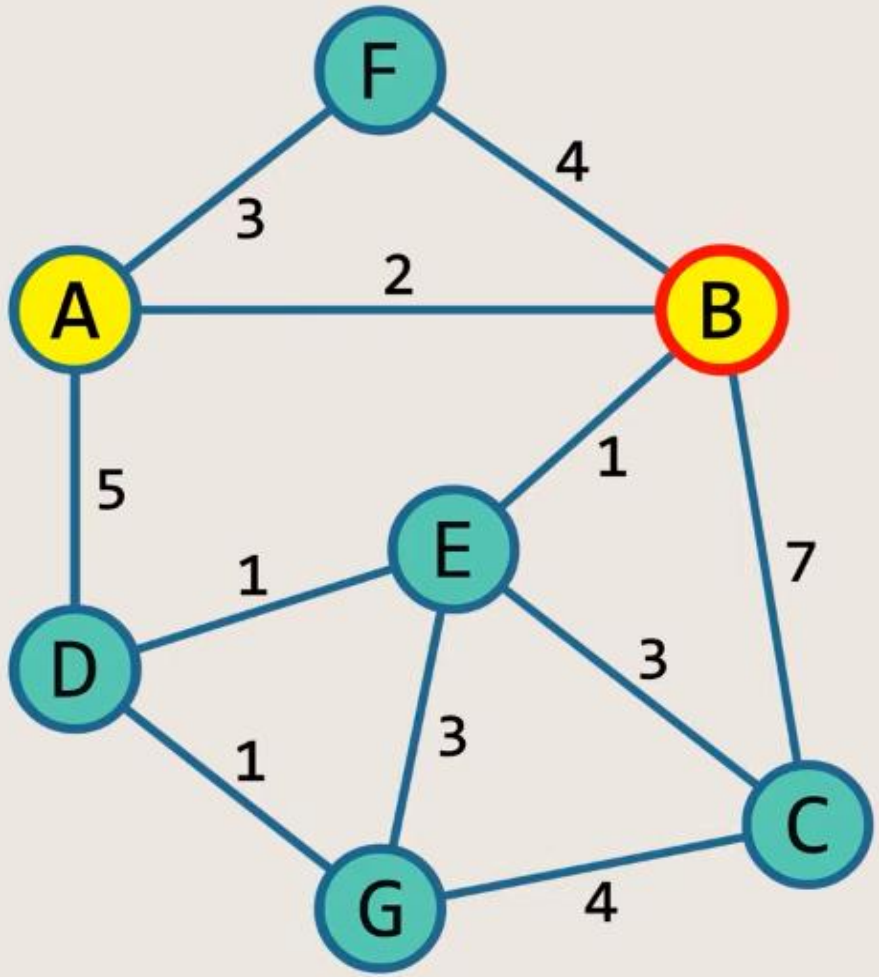
Node	Cost	Previous
A	0	
B	2	A
C	∞	
D	5	A
E	3	B
F	3	A
G	∞	

F,3 D,5 E,3



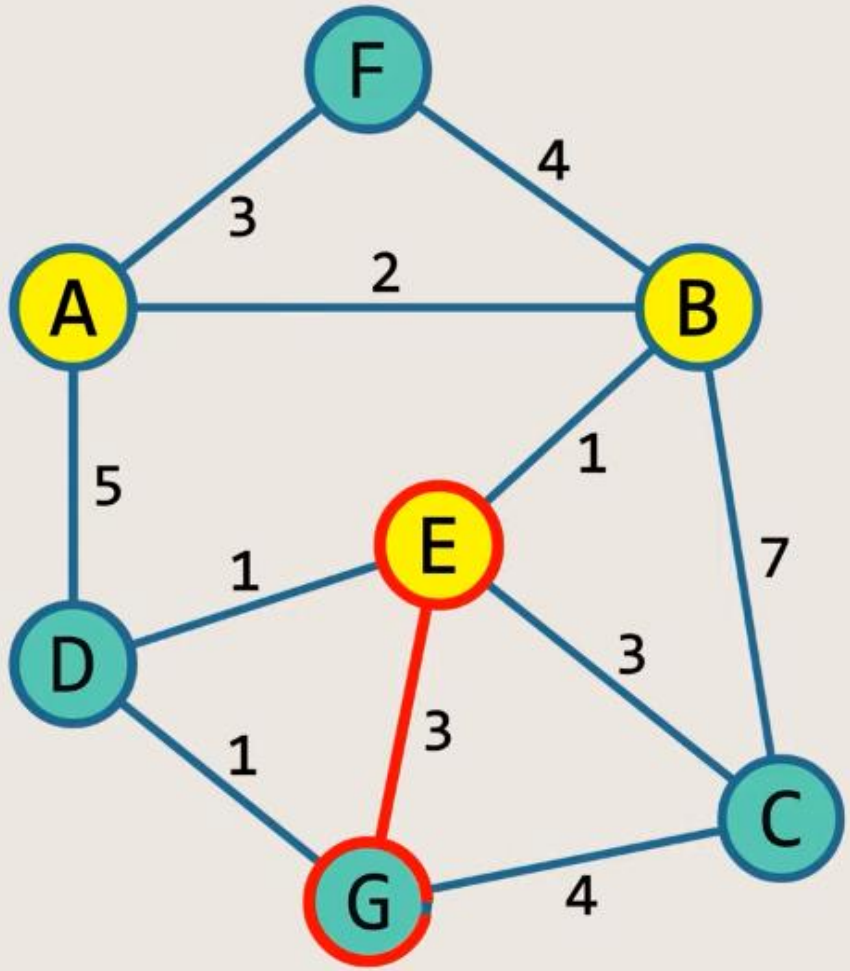
Node	Cost	Previous
A	0	
B	2	A
C	9	B
D	5	A
E	3	B
F	3	A
G	∞	

F,3 | D,5 | E,3 | C,9



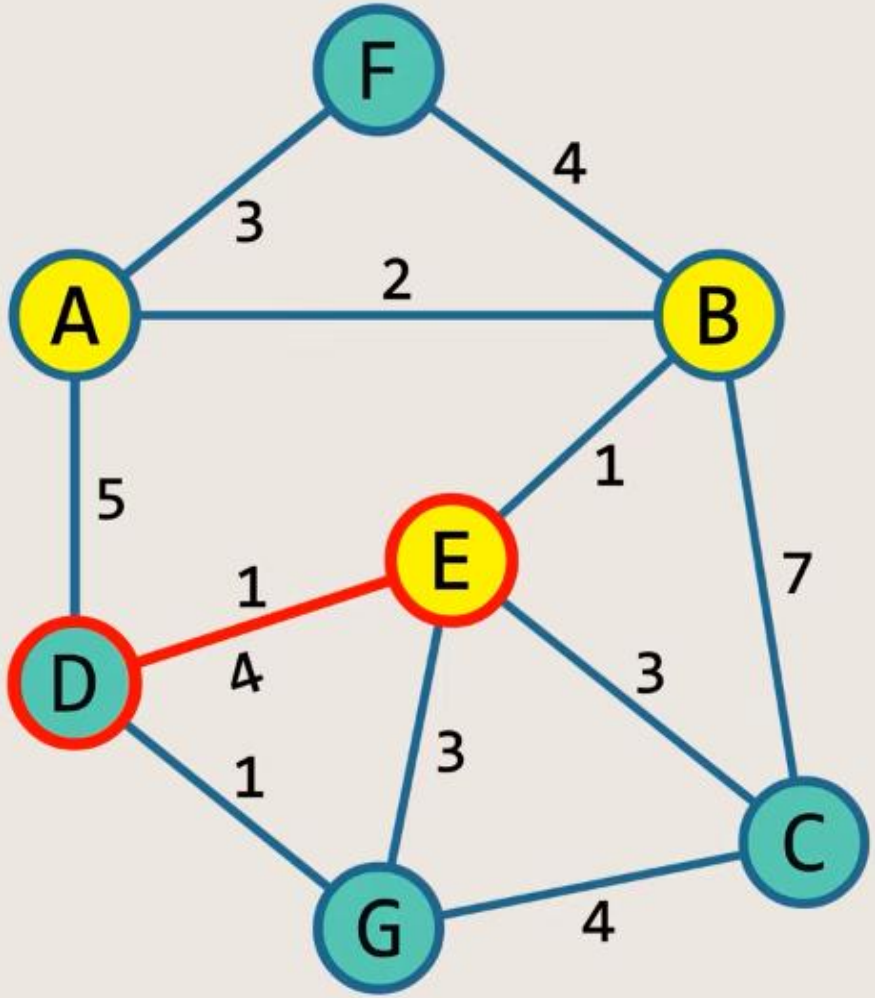
Node	Cost	Previous
A	0	
B	2	A
C	9	B
D	5	A
E	3	B
F	3	A
G	∞	

E,3 | F,3 | D,5 | C,9



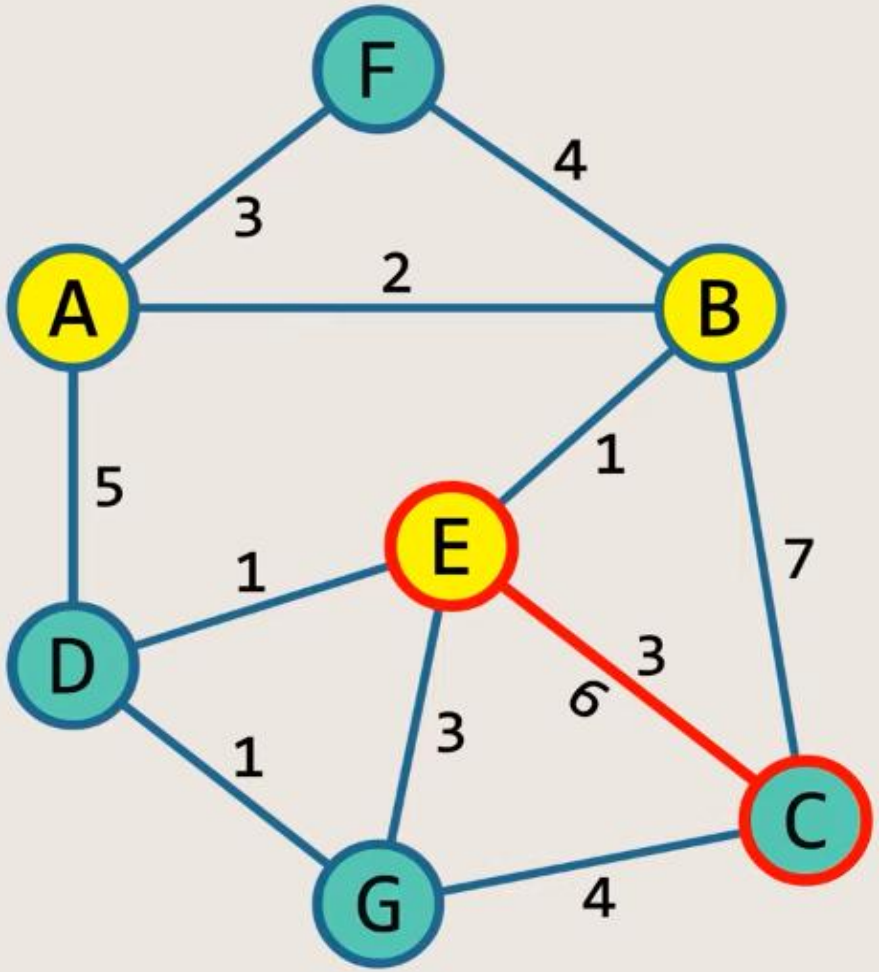
Node	Cost	Previous
A	0	
B	2	A
C	9	B
D	5	A
E	3	B
F	3	A
G	6	E

F,3 | D,5 | C,9 | G,6



Node	Cost	Previous
A	0	
B	2	A
C	9	B
D	4	E
E	3	B
F	3	A
G	6	E





Node	Cost	Previous
A	0	
B	2	A
C	6	E
D	4	E
E	3	B
F	3	A
G	6	E



Bellman-Ford algorithm

https://www.youtube.com/watch?v=lyw4FaxrwHg